

Losing Aid, Losing Ground? The Academic and Career Consequences of Financial Aid Loss: Evidence from the TEXAS Grant

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Abstract

This paper examines the consequences of losing need-based financial aid after the second- and third-year renewal checkpoints of the TEXAS Grant. Using Texas administrative data and a multidimensional difference-in-discontinuities design that exploits joint GPA and credit-completion renewal requirements, I estimate how aid loss affects students' trajectories later in college. Loss of the TEXAS Grant leads to substantial changes in students' aid packages, including a net \$3,500 reduction in total aid even after accounting for a \$1,500 increase in loans. These financial responses are followed by short-run academic disruptions, most notably declines in persistence and enrollment intensity, with the largest effects concentrated among students who narrowly miss only the credit-completion requirement despite otherwise strong academic performance. However, these immediate responses do not translate to impacts on graduation, major choice, or early-career earnings. As the main analysis centers on students who lose aid, I also present reduced-form estimates showing that renewal ineligibility itself lowers immediate persistence, capturing an earlier enrollment response that occurs before aid loss is observed. Overall, the findings highlight how renewal criteria can generate substantial disruption later in college and motivate policy designs that preserve support for students facing temporary setbacks while maintaining clear standards for progress.

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1 Introduction

Higher education in the U.S. is financed through a complex system of financial aid programs, including federal grants, state and institutional scholarships, and student loans. The rising cost of higher education has made financial aid an even more critical tool for supporting college access and affordability. Over the past several decades, tuition has outpaced income growth, making it more difficult for families to cover college costs out of pocket, thereby deepening their dependence on financial aid (Ma et al., 2024). To offset these rising costs, federal and state policymakers have expanded financial aid programs, keeping net cost of attendance relatively stable (NCES, 2023).

While these efforts have mitigated some of the financial burden on students, they have also made financial aid an increasingly central component of the higher education system. For example, in the 2022-23 academic year, 85.2% of first-time undergraduate students received some form of financial aid, averaging \$12,997 in grants and \$7,709 in loans (NCES, 2023). Financial aid is a critical lifeline for many students: 40.7% of undergraduates received a Pell Grant, indicating widespread financial need among college students (NCES, 2023). Despite these investments, the ability of financial aid to keep pace with college costs remains uncertain, particularly as state and institutional budgets tighten, limiting future support for aid programs.¹

Uncertainty at the federal level further complicates the stability of financial aid. The U.S. Department of Education has faced significant challenges in implementing new aid initiatives, as well as delays in grant disbursement and abrupt shifts in loan repayment policies. The Trump administration’s temporary freeze on federal grant aid, which halted funding for various education and research programs pending policy review, underscored the risks posed by unpredictable federal interventions (OMB, 2025). Coupled with changes to Title IV funding mechanisms, such disruptions raise concerns about the reliability of financial aid as a long-term support system.

Administrative hurdles in the financial aid system also create challenges for students, particularly in maintaining eligibility. Recent delays in processing the Free Application for Federal Student Aid (FAFSA) have raised alarms about disruptions to both initial enrollment and ongoing persistence, especially for students most reliant on financial assistance. The complexity of the FAFSA process, combined with uncertainty around eligibility criteria, can discourage students from applying or reapplying for aid—even when they remain eligible (Bettinger et al., 2012; Kofoed, 2017). Research shows that simplifying the financial aid process increases college attendance and retention, underscoring how administrative burdens can function as de facto barriers to access (Castleman and Page, 2016). Students who fail to renew their aid or miss critical deadlines may experience sudden financial shocks, often forcing them to increase work hours, reduce course loads, or drop out entirely.

A particularly consequential source of instability is financial aid renewal requirements, which

¹For example, Indiana’s primary need-based grant, the Frank O’Bannon Grant, will reduce award amounts for most students beginning in fall 2025 (Indiana Commission for Higher Education, 2025), and Missouri’s merit-based Bright Flight Program saw its maximum award decline by 40% in fall 2022 due to budget limitations (Missouri Coordinating Board for Higher Education, 2020).

often require students to maintain a minimum GPA or complete a set number of credits each year. Federal regulations, for example, require students to demonstrate Satisfactory Academic Progress (SAP), typically defined as maintaining at least a 2.0 GPA and completing 67% of attempted credits. These requirements can have significant consequences: more than one-quarter of initial Pell Grant recipients at community colleges lose eligibility due to SAP violations (Schudde and Scott-Clayton, 2016). State programs often impose more stringent criteria, causing even students in good academic standing to lose aid. For instance, the Georgia HOPE Scholarship requires recipients to maintain a 3.0 cumulative GPA, contributing to 35% of initial recipients losing the award by the second renewal checkpoint (GA ORPA, 2024).

Despite extensive research showing that reducing college costs through grants and scholarships increases enrollment, persistence, and completion (Castleman and Long, 2016; Denning et al., 2019), little is known about the consequences of losing financial aid. Given the central role of aid in funding higher education, losing it may have especially disruptive effects. Yet research on the impact of aid loss remains limited, leaving policymakers and institutions without clear guidance on how students respond or which groups are most affected.

In this paper, I investigate the consequences of losing financial aid due to unmet renewal requirements in the Toward EXcellence, Access, and Success (TEXAS) Grant program, examining effects on academic performance, degree attainment, major choice, and early-career earnings. While prior research has focused exclusively on first-year benchmarks, this study shifts the lens to aid loss after the second and third years of college, shedding light on how students respond when financial support is withdrawn later in their academic careers. Further, I estimate the effects of aid loss separately for students who fall short of only the GPA threshold, only the credit threshold, or both, highlighting meaningful heterogeneity in effects across these groups.

Results indicate that the effects of aid loss depend primarily on which renewal criterion is unmet, rather than on whether aid is lost after the second or third year. Losing the TEXAS Grant leads to immediate declines in persistence and enrollment intensity, with especially pronounced responses among students who fail to meet the credit completion requirement despite otherwise strong academic performance. Importantly, these short-run academic disruptions do not translate into systematic effects on timely graduation rates or overall degree completion. Analyses of major choice and early-career earnings likewise reveal little evidence of longer-run impacts. Taken together, the results highlight the importance of renewal design—particularly credit-based requirements—in shaping students’ academic trajectories later in college.

This paper proceeds as follows. Section 2 details the TEXAS Grant program and its renewal criteria. Section 3 reviews the relevant literature and highlights this paper’s key contributions. Section 4 describes the data, while Section 5 outlines the empirical strategy. Sections 6 through 8 present the empirical results, including the main estimates, heterogeneity analyses, and robustness checks. Section 9 concludes by summarizing key findings and discussing implications for the design of financial aid renewal requirements.

2 TEXAS Grant

2.1 Program Overview

The TEXAS Grant is a statewide, need-based aid program that began in the 1999-2000 academic year to provide financial aid to students attending public universities in Texas. To initially qualify for the TEXAS Grant, one must be a Texas resident with financial need and enroll at least three-quarter time within 16 months of graduating high school.² Unlike many other state programs, the TEXAS Grant does not require a minimum high school GPA or SAT/ACT score for eligibility.

The TEXAS Grant operates as a last-dollar scholarship, covering any remaining tuition and required fees after other non-loan federal, state, and institutional aid has been applied. While the program helps reduce the direct cost of attendance, it does not cover living expenses such as room, board, or transportation. Thus, many recipients still rely on loans or other aid to meet these needs. The average annual award is approximately \$6,000, accounting for nearly one-third of a recipient's total financial aid package. Students may receive up to five years of funding if they continue to meet program requirements.

The scale and reach of the TEXAS Grant underscore its importance in the state's higher education landscape. In the 2020–21 academic year, nearly 83,000 students—about one in six undergraduates at public universities—received TEXAS Grant funding, totaling over \$400 million in state disbursements. Since its inception, the grant has supported hundreds of thousands of students, with roughly 13% of in-state graduates receiving the award at some point in their college careers. Funding is distributed across all four-year public universities, including those in the University of Texas and Texas A&M systems. More than half of recipients (54%) are Hispanic, and a majority (56%) have an expected family contribution (EFC) of \$0, highlighting the program's focus on supporting low-income and historically underserved students.

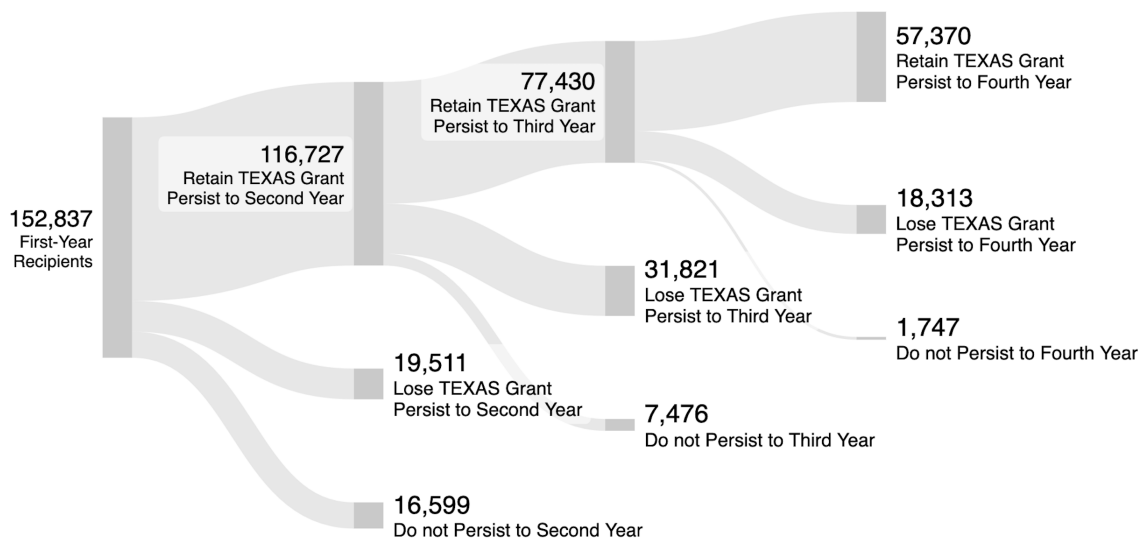
2.2 Renewal Requirements and Aid Loss

To continue receiving the TEXAS Grant, students must meet their institution's SAP standards after their first year. These criteria can vary across universities, making first-year renewal standards non-uniform. Beginning in the second year, however, renewal requirements are standardized statewide: all students must maintain a 2.5 cumulative GPA and complete at least 24 credit hours each academic year, with falling below either requirement placing students at risk of losing the grant. These renewal criteria are unique among Texas aid programs and differ from federal eligibility rules.

As shown in Figure 1, a significant number of students failed to meet these benchmarks. Among the 152,837 students who entered between fall 2011 and fall 2018 and received the TEXAS Grant in their first year, 136,238 (89.1%) returned for a second year, but 19,511 (14.3%) lost the grant despite

²Students must also be registered for the Selective Service and never have been convicted of a felony or any offense involving a controlled substance. While 95% of recipients enter from high school, students can also qualify if they have earned an associate's degree and pursue a bachelor's within a year of completion; enroll after an honorable discharge from the military; or satisfy additional requirements after beginning their bachelor's degree with the Texas Educational Opportunity Grant. This paper only considers students entering immediately after high school.

persisting. Attrition at later renewal points was even higher: 31,821 of the 109,251 students (29.1%) who returned for a third year did not renew. By the fourth year, 18,313 of the remaining 75,683 recipients (24.1%) failed to renew. Among students who lost the grant due to unmet benchmarks, 43% fell short on both GPA and credit requirements, 35% fell below only the credit threshold, and 22% fell below only the GPA requirement.³



Notes: This figure shows the flow of initial TEXAS Grant recipients in the fall 2011–2018 entering cohorts as they exit college and the program over time.

Figure 1. Flow of TEXAS Grant Recipients

These statistics highlight the prevalence of aid loss in the TEXAS Grant program. Among those who remained enrolled, nearly 70,000 students in these eight cohorts failed to renew their grants, resulting in substantial repercussions to their financial aid packages. For instance, a student who did not renew the grant after their second year lost \$6,000 per year, or \$18,000 if they graduate in five years. Many of these students cannot afford to attend college without this support, forcing them to replace the lost aid, reduce their enrollment, or drop out.

3 Literature Review and Contributions

3.1 Literature Review

This paper primarily contributes to the literature on the consequences of financial aid loss. When students lose grant aid, they often turn to alternative funding sources, with loans being the most immediate and accessible option. This shift in aid composition can affect persistence, graduation, and financial and academic trajectories. To provide context for why aid loss may impact students,

³Some students lose aid despite meeting both criteria. I exclude these cases from this breakdown, as the reason for aid loss is not observable in the data.

this section first reviews prior research on how financial aid receipt affects student outcomes as well as selected work on student debt and career choices—which I explore as a potential downstream consequence of aid loss—before turning to the existing literature on aid loss itself.

3.1.1 Financial Aid, Debt, and Student Outcomes

Research consistently finds that lowering the net cost of college improves enrollment, persistence, and completion (Nguyen et al., 2019). Need-based grants, such as the federal Pell Grant, have a particularly strong impact on enrollment and degree attainment (Goldrick-Rab et al., 2016; Castleman and Long, 2016; Denning et al., 2019; Denning, 2019). Prior work on the TEXAS Grant itself finds mixed evidence of post-enrollment effects: Villarreal (2018) documents positive impacts on degree completion and some credit accumulation, while Montenegro (2020) and Galperin (2024) find limited effects on academic outcomes. In contrast, merit-based aid primarily influences where students enroll rather than whether they attend college (Mayer et al., 2015). While it may shift enrollment choices, merit aid does not significantly affect student achievement (Monks, 2009; Sjoquist and Winters, 2015; Angrist et al., 2022).

The effects of financial aid extend beyond college, as financial constraints shaped by aid availability can influence students’ academic paths, career choices, and labor market outcomes. Stater (2011) finds that a higher net cost of attendance increases the likelihood of students selecting professional majors while decreasing their likelihood of majoring in the humanities and certain sciences. Reducing student loan burdens has also been shown to increase the likelihood that graduates pursuing lower-paying public service and nonprofit careers, whereas higher debt levels push students toward higher-paying private-sector jobs (Rothstein and Rouse, 2011). Building on this, Hampole (2023) examines the staggered implementation of more than 20 “no-loan” policies across over 700,000 graduates, showing that students with higher debt levels choose majors with higher initial earnings, but lower lifetime earnings. This effect is particularly pronounced among low-income students, who are more sensitive to debt burdens in their decision-making.

Several studies have also established a clear link between student debt and mental health, raising concerns that loan reliance can impose psychological burdens as well. Research has associated high debt burdens with increased stress, anxiety, and depression (Selenko and Batinic, 2011; Sweet et al., 2013; Hojman et al., 2016). More specifically, student debt has been shown to negatively impact mental well-being (Walsemann et al., 2015; Despard et al., 2016), which may in turn influence academic performance, persistence, and career choices (Herzog, 2018; Destin and Svoboda, 2018; Baker and Montalto, 2019). These effects may be especially pronounced for students who unexpectedly lose grant aid and must shift to loans, potentially exacerbating stress and derailing their educational plans. This body of research highlights the broader consequences of debt-financed higher education, suggesting that financial aid policies should consider not only students’ ability to pay but also their mental health and long-term economic security.

Taken together, this literature suggests that losing grant aid midstream may be particularly disruptive: unlike students who never receive aid, those who lose it face an unexpected financial

shock after investing in college. This shock not only reduces affordability but may also alter expectations, forcing students to adjust by borrowing more, reducing course loads, withdrawing, or changing majors.

3.1.2 Financial Aid Loss and Student Outcomes

While the literature on the impacts of financial aid receipt is extensive, relatively little research examines the consequences of losing it. Recent evidence suggests that students value the stability of financial support: [Dynarski et al. \(2021\)](#) show that a guaranteed four-year aid package covering tuition and fees—without requiring annual reapplication—increases both applications and enrollments, suggesting that students place a premium on predictability in financial support. Related work suggests that changes in financial aid status may be more consequential than the initial level of aid a student receives, as fluctuations in support can significantly alter students’ educational trajectories ([Bettinger, 2004](#)). In this sense, losing aid may not only reverse the positive effects of initial aid receipt—undoing the very gains that aid programs aim to generate—but may also impose new barriers to persistence and completion. This interpretation is consistent with the loss aversion literature in behavioral economics: once anchored in expectations of continued support, students may respond more strongly to aid removal than to its initial receipt ([Kahneman et al., 1991](#)).

In the small body of empirical work on aid loss, [Henry et al. \(2004\)](#) examines the Georgia HOPE Scholarship, which covers full tuition, books, and mandatory fees for in-state students with at least a 3.0 high school GPA. To retain the scholarship, students must maintain a 3.0 cumulative GPA and attempt at least 30 credits per year. His findings indicate that students who lost eligibility had similar credit accumulation, GPAs, persistence, and four-year graduation rates to those who narrowly missed qualifying upon entry. However, students who retained the scholarship throughout college had significantly better outcomes than both groups.

Further evidence on the effects of aid loss comes from [Carruthers and Özek \(2016\)](#)’s examination of Tennessee’s HOPE Scholarship. This program covers approximately 70% of tuition and fees for up to five years for Tennessee high school graduates who met either the 3.0 high school GPA requirement or scored at least a 21 on the ACT. Continued eligibility required maintaining a 2.75 cumulative GPA after 24 attempted credit hours and a 3.0 cumulative GPA at subsequent checkpoints (48, 72, and 96 credit hours). Using a regression discontinuity design (RDD) at the first GPA renewal threshold, the authors find that students who lost eligibility became slightly detached from college, as reflected in a 14 cent increase in earnings per dollar of aid lost. However, losing the scholarship did not appear to affect overall persistence or graduation rates.

More recent work by [Cummings et al. \(2022\)](#) reexamines the consequences of losing the Tennessee HOPE Scholarship at the same checkpoint for cohorts entering between fall 2011 and fall 2014. Losing the scholarship meant a reduction of roughly \$2,000 in aid per semester, which had differential consequences across racial and socioeconomic groups. Students who lost aid were more likely to stop out if they were White and higher income, while Black students had higher community college transfer rates and lower on-time degree completion compared to their peers who retained

the scholarship.

Related evidence from the community college sector comes from [Columbus et al. \(2025\)](#), who examine the consequences of losing eligibility for the Tennessee HOPE scholarship due to the cumulative GPA renewal requirement. The authors show that ineligibility results in higher rates of stopout, fewer transfers to four-year institutions, and lower rates of associate degree completion. These findings highlight that the loss of eligibility for financial aid can meaningfully alter students' enrollment trajectories even outside four-year settings.

Finally, and most relevant to this paper, is [Jones et al. \(2021\)](#)'s research on changes to the Georgia HOPE program. Beginning in 2011, the HOPE Scholarship was reduced to cover only 90% of tuition, eliminating funding for books and mandatory fees. In its place, the newly created Zell Miller Scholarship covered these costs in full but imposed stricter eligibility requirements: a 3.7 high school GPA, a 1200 SAT (or 26 ACT) score, and a 3.3 cumulative college GPA to maintain eligibility. Continuing students were not grandfathered into the new program, so students already enrolled who did not meet the higher thresholds experienced a modest 10-15% reduction in aid based on pre-college academic criteria rather than contemporaneous college performance. To study these changes, the authors employ a multidimensional RDD that leverages the new high school GPA and SAT/ACT requirements to estimate the effects of this partial aid reduction for cohorts entering college no more than two years before the change. Unlike traditional RDD approaches that rely on a single cutoff, this method accounts for multiple eligibility thresholds simultaneously, allowing for a more precise identification of students' varying exposure to aid reductions. They find no significant impact on persistence or graduation rates, suggesting that higher-achieving students may be less sensitive to aid loss or that the reduction in aid was not substantial enough to disrupt educational trajectories.

3.2 Contributions

This paper makes several contributions to the aid loss literature within the broader field of financial aid research. First, it examines a large-scale need-based financial aid program, whereas existing works focus on merit-based programs. As need-based recipients often have fewer resources to replace lost aid, the consequences of losing support may be more severe, affecting not only continued enrollment but also academic progress, performance, and major choice. These consequences may be especially pronounced in the context of the TEXAS Grant, where students lose access to approximately \$6,000 per year in aid with no opportunity to regain eligibility once it is lost—a structure that makes the program higher stakes than others studied in the existing literature.

Second, the TEXAS Grant conditions renewal on both cumulative GPA and credit completion thresholds, while merit-based programs typically assess GPA at fixed credit checkpoints but do not require students to complete a minimum number of credits each year. As a result, TEXAS Grant recipients must maintain both satisfactory grades and steady academic progress, whereas students in merit-based programs generally need only be concerned with their GPA. This paper exploits this structure by estimating the overall effect of aid loss from failing to meet any renewal requirement,

as well as separate effects based on whether students fell below only the GPA threshold, only the credit threshold, or both. Rather than treating these distinctions as supplementary, the analysis places them at the center, recognizing that GPA- and credit-based aid loss reflect different academic challenges, student circumstances, and expectations around continued eligibility. Importantly, aid loss among students below only the credit threshold occurs among students otherwise in good academic standing, offering new evidence that even higher-achieving students respond to losing grant aid. These students are also the most likely to complete their degree if they remain enrolled, making their attrition particularly costly for both students and policymakers. By isolating effects for students below only one threshold, the study provides new insight into how the interaction between multiple renewal criteria shapes student responses to aid loss, offering important implications for financial aid policy and institutional support strategies.

Further, this paper provides the first evidence on full need-based aid loss at later renewal checkpoints—specifically, after the second and third years of college—offering insight into how the consequences may differ for students further along in their academic careers and the implications for policies aimed at supporting persistence and degree completion. Losing aid at this stage is particularly costly, both for students, who have already invested significant time and credits, and for states, which have already subsidized much of their education. The consequences may also differ from first-year aid loss, since students later in their academic careers have fewer semesters to recover from setbacks and may face different expectations about their likelihood of graduating. Students affected at these later stages also differ from those at initial renewal checkpoints, as they have already persisted longer and tend to be higher achieving and more likely to graduate, providing insight into how aid loss impacts a distinct group of recipients.

The final contribution of this paper is its expanded scope of outcomes examined in response to aid loss. In addition to persistence and graduation, I analyze students’ academic adjustments (e.g., credit accumulation, GPA, and major switching), financial coping strategies (e.g., shifts in grants, loans, work-study, and outside earnings), and early-career labor market outcomes. By tracking these responses over multiple years, the analysis offers the most comprehensive picture to date of how students navigate the loss of grant aid. Together, these contributions offer a more complete picture of how and why financial aid loss affects students, highlighting the importance of program structure, timing, and student circumstances. This broader understanding is essential for designing aid policies that promote not just access, but long-term student success.

4 Data

The data for this paper come from the Houston Education Research Center, which houses longitudinal data provided by the Texas Education Agency (TEA), the Texas Higher Education Coordinating Board (THECB), and the Texas Workforce Commission (TWC). These records track students who attended both high school and a public university in Texas, linking their postsecondary enrollment, academic performance, and employment histories.

The postsecondary data include student admissions information, annual financial aid records, semester-by-semester courses, grades, and majors, as well as graduation outcomes. Admissions data include student and family demographics, such as age, gender, race/ethnicity, parental education, and family income range. The financial aid records contain an indicator for TEXAS Grant receipt, along with all other sources and amounts of aid awarded to each student. These data allow me to examine how students adjust their aid packages after losing the TEXAS Grant by disaggregating changes across grants, loans, and work-study. All financial aid amounts are expressed in 2021 dollars using the Bureau of Labor Statistics’ College Tuition and Fees Price Index (Series ID: CUUR0000SEEB01).⁴

The semester-by-semester course and grade data provide the information needed to calculate cumulative GPA and credit accumulation, which determine whether a student meets TEXAS Grant renewal requirements. Students’ declared major(s) each semester are identified using 6-digit Classification of Instructional Programs (CIP) codes, while graduation records capture graduation dates, majors, and degrees earned. Because the dataset covers all Texas public universities, I can track students who transfer and graduate within these schools; however, transfers out of state or to private institutions are not observed.

Employment records contain quarterly wage reports from Texas employers, detailing both earnings and North American Industry Classification System (NAICS) employment sector. I use these data to examine changes in earnings patterns in response to aid loss during enrollment and after leaving or graduating. All earnings data are adjusted to 2021 dollars using the Consumer Price Index for All Urban Consumers (Series ID: CUUR0000SA0). These records do not fully capture earnings from informal employment, gig work, or self-employment, which can be significant sources of income for students.

To align with the availability of semester-by-semester course and grade data required to track GPA and credit accumulation, this analysis uses data spanning from fall 2011 through spring 2022. I restrict analysis to cohorts entering between fall 2011 and fall 2018, ensuring that all students are observed for at least four years. Since the TEXAS Grant is only available to in-state students, I further limit to students who attended high school in Texas. Additional restrictions, described in the methodology section, further refine the analytic sample.

5 Methodology

5.1 Framework

As the first-year TEXAS Grant renewal thresholds vary across universities, I focus on the second- and third-year checkpoints, when all recipients must maintain a 2.5 cumulative GPA and complete 24 credits annually to retain eligibility. Students become ineligible if they fall short of either requirement, and those attempting fewer than 24 credits automatically lose eligibility regardless of

⁴The final academic year included in the financial aid data is 2021-22, making 2021 the most appropriate base year for inflation adjustments.

GPA. To ensure that eligibility status is not mechanically driven by endogenous enrollment choices, I restrict the sample to students who attempted at least 24 credits in the renewal year.⁵ This restriction reflects the policy rule itself and ensures that variation in eligibility arises from academic performance rather than mechanically from under-enrollment. I estimate overall effects of aid loss using both GPA and credits as running variables in a multidimensional framework, and I obtain threshold-specific effects by reducing to a single running variable while restricting the sample to students who satisfied the other requirement.

Although the multidimensional RDD framework provides an estimate of the impact of aid loss, it does not isolate the causal effect of losing financial aid from other academic disruptions. Since all students attempted at least 24 credits, falling below the credit requirement necessarily implies at least one course failure. Course failure may carry its own academic or psychological consequences that influence subsequent enrollment, major choice, and graduation outcomes—effects that an RDD would not disentangle. Simply comparing students above and below the renewal thresholds would therefore risk conflating the effects of aid loss with those of course failure.

To address this concern, I incorporate non-TEXAS Grant recipients as a comparison group in a difference-in-discontinuities (diff.in-disc.) framework. These non-recipients include both students who received other forms of financial aid and those who received none. Given that approximately three-quarters of financial aid recipients receive the TEXAS Grant, restricting the comparison group to only non-TEXAS Grant aid recipients would result in a much smaller comparison group, particularly when examining outcomes later in students’ academic careers. Including all students who faced similar academic challenges but were not TEXAS Grant recipients allows me to net out the effects of course failure and isolate the causal impact of financial aid loss.⁶

I further restrict the analysis to students with no prior course failures and who were enrolled full-time in all semesters leading up to the renewal checkpoints. These restrictions ensure that students have similar academic performance and enrollment intensity prior to the point at which renewal eligibility is determined. For the second-year benchmark, this means students passed at least 24 credits with no failed courses in their first year and attempted at least 24 credits in their second. The third-year sample is similarly defined based on successful completion of credits in the first two years and attempted credits in the third. As a result, differences observed after the renewal checkpoint are not driven by prior academic setbacks or enrollment choices.

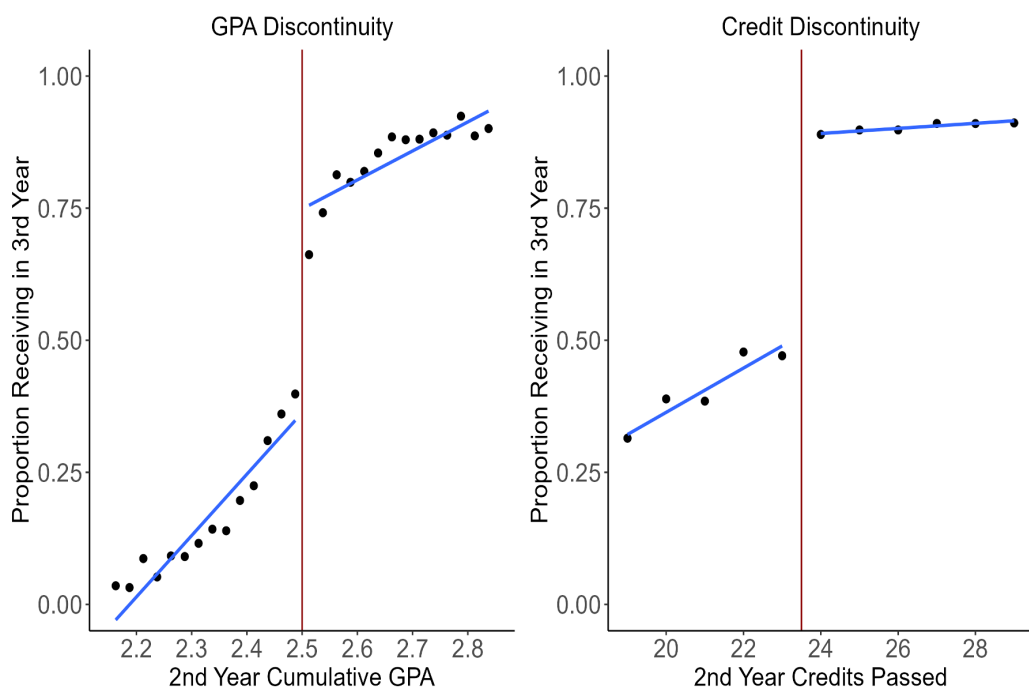
Although eligibility is based on clearly defined thresholds, actual grant renewal is not deterministic. Not all TEXAS Grant recipients who fell below a renewal threshold ultimately lost the grant in the subsequent year. Figure 2 illustrates the proportion of students who continued to receive the TEXAS Grant in their third year based on each second-year renewal requirement.⁷ Among

⁵This count includes only courses for which a student received a grade, including courses marked as incomplete. I exclude courses from which a student has withdrawn, as I do not observe the timing or reason for the withdrawal.

⁶Non-recipients are subject to the same sample restrictions as recipients, including attending high school in Texas. Out-of-state students face a significantly higher financial burden, further justifying this restriction. When analyzing financial aid outcomes, I assign non-recipients who did not receive aid a value of zero for all aid components (grants, loans, and work-study) to ensure consistency in comparisons.

⁷It is not straightforward to visualize a multidimensional first stage. I therefore present first stages separately by

those with cumulative GPAs below the threshold, approximately 35% of those within 0.1 points still received the grant, but this share falls to 5–10% for those further below. In contrast, 65–90% of students who met the GPA requirement renewed. For the credit benchmark, 25–50% of students below the threshold retain the grant, compared to over 85% among those who met it. Similar patterns hold for third-year GPA and credit benchmarks, as shown in Figure A.1.



Notes: This figure shows the first-stage for each renewal threshold separately at the second-year benchmark.

Figure 2. First Stage – Year 2 Benchmark

One reason some students continued receiving the grant despite failing to meet renewal requirements is the appeal process for aid loss. Students who can demonstrate that they experienced hardships during the year they lost eligibility may retain the grant.⁸ Data on appeals is not available, making it difficult to assess the extent to which appeals contributed to continued grant receipt for ineligible students.

There is also no clear evidence that certain universities systematically applied renewal criteria more leniently or used discretionary funds to extend aid to ineligible students. However, there is a strong relationship between proximity to a renewal threshold and the likelihood of continuing to receive the grant: students who narrowly missed the GPA or credit requirements were more likely to retain the TEXAS Grant than those further below a cutoff. Students who retained the grant despite falling short of renewal requirements also differed from those who lost it—they tended to

GPA and credit thresholds, conditional on being above the other threshold, to illustrate the pattern of imperfect compliance along each dimension.

⁸Students must submit a letter of explanation documenting their hardship, which may include severe illness, responsibility for the care of another individual, or loss of a family member, among others.

have higher GPAs and were less likely to be male or Black.⁹ This may stem from the appeals process, institutional discretion, or unobserved administrative factors, with the GPA pattern in particular suggesting that stronger academic performance may have increased the likelihood of leniency. I return to the implications of this pattern later in the methodology section.

Since the renewal requirements do not perfectly predict continued receipt of the TEXAS Grant, I use a fuzzy multidimensional diff.-in-disc. design. I extend the framework of [Choi and Lee \(2018\)](#) and [Jones et al. \(2021\)](#) to accommodate both the fuzziness in treatment assignment and the inclusion of non-recipients. The first-stage uses the following equation:

$$\begin{aligned}
(Lose_{i,t+1} \times Recipient_{it}) = & \pi_0 + \pi_1(1\{GPA_{it} < 2.5 \text{ or } Credits_{it} < 24\} \times Recipient_{it}) + \\
& \gamma Recipient_{it} + f(GPA_{it}, Credits_{it}) + \\
& [Recipient_{it} \times f(GPA_{it}, Credits_{it})] + X_{it} + W_{i,t-1} + \\
& Major_{i,t=1} + Institution_i + Cohort_i
\end{aligned} \tag{1}$$

where $(Lose_{i,t+1} \times Recipient_{it})$ is the endogenous treatment variable that captures the effect of losing the TEXAS Grant among prior recipients. The instrument is the interaction $1\{GPA_{it} < 2.5 \text{ or } Credits_{it} < 24\} \times Recipient_{it}$, which captures exogenous variation in grant loss induced by falling below a threshold. X_i is a vector of student demographic characteristics including gender, age, race/ethnicity, and family income. $W_{i,t-1}$ is a vector of pre-renewal academic outcomes that includes cumulative GPA, credits attempted and passed, and the number of math, science, social science, and English credits completed through the year prior to the renewal checkpoint. Fixed effects for 2-digit CIP entry major, institution, and cohort are also included.

The function $f()$ accounts for the multidimensional nature of the running variables and is specified as:

$$\begin{aligned}
f() = & GPA\ Below_{it} + GPA\ Above_{it} + Credits\ Below_{it} + Credits\ Above_{it} + \\
& (GPA\ Below_{it} \times Credits\ Below_{it}) + (GPA\ Above_{it} \times Credits\ Above_{it})
\end{aligned} \tag{2}$$

where $GPA\ Below_{it}$ and $Credits\ Below_{it}$ are forcing variables equal to centered GPA and centered $Credits$ if they are below their respective threshold, otherwise 0. $GPA\ Above_{it}$ and $Credits\ Above_{it}$ are defined analogously for values above their respective threshold. I center GPA at 2.5 and $Credits$ at 24, so negative values indicate ineligibility and positive values eligibility.

These forcing variables serve two purposes. First, they provide flexible controls for smooth trends in academic performance just above and below each threshold, ensuring that the estimated effects are not driven by underlying differences in GPA or credit completion. Second, by incorporating both GPA and credit running variables, the specification captures the joint determination of eligibility. The inclusion of interaction terms between GPA and credit thresholds further accounts for students

⁹See Appendix Table [A.1](#) for characteristics of compliers and non-compliers who fell below at least one threshold at the second-year benchmark. Comparisons are presented by specification, introduced later. Compliance is highest in the Diagonal and GPA specifications, while differences are most pronounced in the Credit specification.

who are just above one threshold but just below the other, allowing for more precise estimation of the effect of aid loss across different academic profiles.

First-stage estimates show that falling below a renewal threshold(s) reduces the probability of continued grant receipt by 44 to 49 percentage points (pp) depending on the bandwidth, confirming the strength of the instrument. My second-stage equation is then specified as follows:

$$\begin{aligned}
Y_{i\tau} = & \alpha + \beta(\widehat{\text{Lose}_{it+1} \times \text{Recipient}_{it}}) + \gamma_1(1\{GPA_{it} < 2.5 \text{ or } Credits_{it} < 24\}) + \\
& \gamma_2\text{Recipient}_{it} + f(GPA_{it}, Credits_{it}) + \\
& [\text{Recipient}_{it} \times f(GPA_{it}, Credits_{it})] + X_i + W_{i,t-1} + \\
& Major_{i,t=1} + Institution_i + Cohort_i + \epsilon_{it}
\end{aligned} \tag{3}$$

where $Y_{i\tau}$ is an outcome relating to subsequent financial aid, academic outcomes, graduation, major changes, or earnings, with $\tau \geq t+1$, and all other variables are as previously defined. The coefficient of interest, β provides the local average treatment effect (LATE) of losing the TEXAS Grant for recipients near at least one renewal threshold.

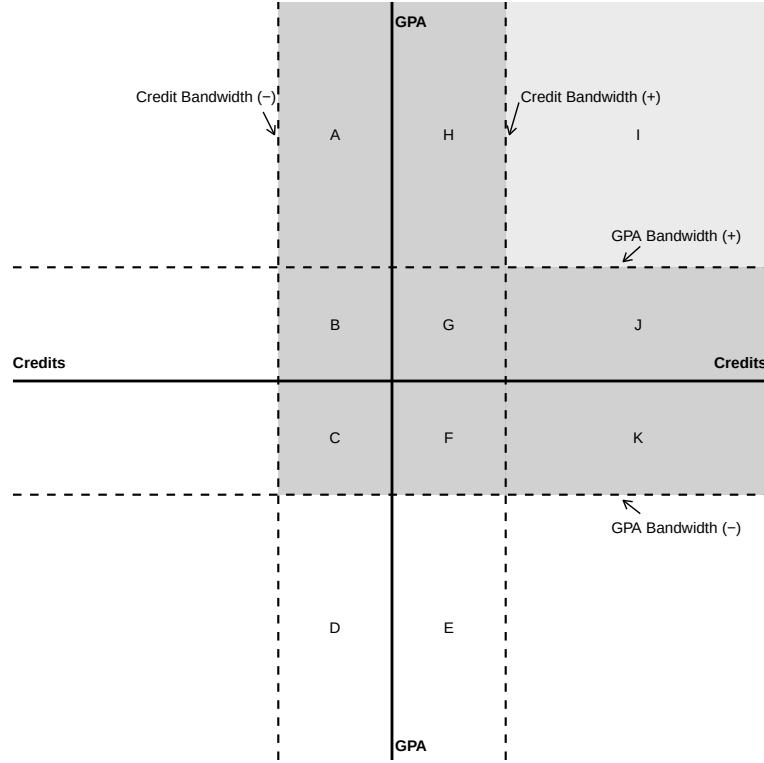
It is important to clarify that I define treatment based on whether a student loses the grant in the subsequent year ($t+1$). For the second-year benchmark, students who lose the grant after their third or fourth year remain in the analysis but are not classified as “treated” at this stage. That is, the design compares TEXAS Grant recipients (and non-recipients) who lost the grant after the second year to those who retained it. This avoids comparing students who lost the grant after two years to those who never lost, which would overstate the effects on outcomes like graduation, given that continued receipt is conditional on persistence—which is itself strongly correlated with degree completion.

Following Jones et al. (2021), I select bandwidths for each dimension separately using the optimal bandwidth procedures from Calonico et al. (2014). The resulting bandwidths are 5 credits and 0.328 GPA points for the second-year benchmark and 5 credits and 0.333 GPA points for the third-year benchmark.¹⁰ My primary, or “Full”, specification includes students close to either cutoff, as shown by the darker shaded region in Figure 3, which encompasses regions A, B, C, F, G, H, J, and K. Students in regions A, B, C, F, and K fail to meet at least one of the renewal requirements, making them ineligible for continued TEXAS Grant receipt.

Limiting the analysis to only students close to one of the renewal requirements covers more than one-quarter of all initial TEXAS Grant recipients who entered between fall 2011 and fall 2018. Because renewal status is observed in the year following each benchmark, the analysis conditions on students returning for a third (or fourth) year. For example, the second-year benchmark sample includes 41,926 of the 109,251 (38.4%) initial recipients who remained enrolled at least three years.

The first two columns of Table 1 compare all initial TEXAS Grant recipients to those included in the second-year benchmark analysis. Recipients in the analysis were slightly younger at their time of

¹⁰Optimal bandwidths are nearly identical if I do not condition on being above the other renewal threshold. For the second-year benchmark, the bandwidths are 5 credits and 0.349 GPA points; for the third-year, they are 5 credits and 0.348 GPA points.



Notes: This figure illustrates the multidimensional design. The horizontal dashed lines represent the GPA bandwidth, while the vertical dashed lines indicate the credit bandwidth. Students far from both cutoffs are excluded—either far above both (I) or far below both (D, E, and all white areas). The included regions for each specification are as follows:

Full: A,B,C,F,G,H,J,K; **Diagonal:** C,G; **Credit:** A,B,G,H; **GPA:** F,G,J,K

Figure 3. Bandwidths and Specification Illustration

application (18.03 vs. 18.08) but were otherwise similar in gender and racial/ethnic composition to the full population of recipients. Since the sample excludes early-leavers, these students accumulated more first-year credits (28.07 vs. 24.44), had higher first-year GPAs (3.07 vs. 2.71), and graduated at higher rates (88% vs. 66%) than all recipients.

The third column of Table 1 provides descriptives of non-TEXAS Grant recipients in the analysis. While non-recipients were similar in age, their racial/ethnic composition differed markedly from recipients. Specifically, a much larger share of non-recipients were white (54% vs. 18%), while a smaller share were Hispanic/Latino (26% vs. 55%), reflecting broader trends in need-based financial aid receipt. First-year credit accumulation is comparable between recipients and non-recipients, though non-recipients had slightly higher first-year GPAs (3.18 vs. 3.07) and graduated at somewhat higher rates (92% vs. 88%).

Although some observable differences exist between recipients and non-recipients, this does not necessarily pose a threat to identification. What matters is whether both groups exhibit similar patterns at the thresholds, ensuring that any observed discontinuities in outcomes are driven by aid loss rather than differential changes in observable characteristics. This assumption is tested in

Variable	All Recipients	Recipients in Analysis	Non-Recipients in Analysis
	(1)	(2)	(3)
Age	18.08	18.03	18.06
Male	0.42	0.40	0.45
White	0.18	0.18	0.54
Black	0.17	0.16	0.08
Hispanic/Latino	0.55	0.55	0.26
Year 1 Credits Passed	24.44	28.07	28.36
Year 1 Cumulative GPA	2.71	3.07	3.18
Graduate	0.66	0.88	0.92
Students	152,837	41,926	140,219

Notes: This table reports descriptive statistics for three groups. Column (1) includes all students who ever received the TEXAS Grant. Column (2) restricts to second-year TEXAS Grant recipients who fall within the optimal bandwidth of at least one second-year renewal threshold and persist to a third year. Column (3) includes non-recipients satisfying the same restrictions.

Table 1. Descriptive Statistics of Recipients and Non-Recipients

later covariate balance checks.

5.2 Effects by Renewal Threshold

The Full specification contains students with a wide range of academic performance, even among those who lose the TEXAS Grant. For instance, a student in region A might have failed a one-credit course while maintaining a 3.5 cumulative GPA, while a student in region K could pass 30 credits but fall just short of a 2.5 cumulative GPA. Because students differ not only in their academic performance but also in the reasons they lose aid, the consequences of aid loss are likely to differ across groups.

Students with stronger academic records may be more likely to persist and have access to alternative financial resources, such as institutional aid or private scholarships. However, they may also perceive their aid as secure, making the loss more unexpected and potentially more disruptive. This dynamic is relevant for students who fall short of the credit requirement but maintain a strong GPA. In contrast, students who lose aid due to falling below the GPA requirement may already be experiencing academic challenges that place them at higher risk of stopping out. For these students, aid loss could either exacerbate existing setbacks or have a smaller marginal impact if it is anticipated and perceived as part of broader academic difficulties. Finally, students who fall below both thresholds may face the most acute risks: they are academically vulnerable on multiple dimensions, and the loss of aid may serve not only as a financial shock but also as a confirmation of limited academic progress.

These academic struggles are not unique to TEXAS Grant recipients—non-recipients experience similar challenges by design. However, only recipients face the additional consequence of losing

financial aid when those struggles cross a renewal threshold. My empirical strategy is designed to isolate this interaction: whether and how the consequences of academic setbacks differ when they also trigger the loss of financial support.

To examine whether the reason for aid loss moderates its effects, I segment the analysis into three groups based on which threshold(s) students fall below. Each student falls into one or more of the following categories:¹¹

1. **Credit Discontinuity.** The first group consists of students near the credit threshold who exceed the GPA requirement, represented by regions A, B, G, and H in Figure 3. Specifically, this includes students who passed between 19 and 29 credits—corresponding to the optimal 5-credit bandwidth around the 24-credit threshold—while maintaining a cumulative GPA of 2.5 or higher. These regions account for 83.8% of the second-year sample and 86.6% of the third-year sample, but include only 24.9% of second-year recipients and 53.4% of third-year recipients who lose the TEXAS Grant. As all students meet the GPA requirement, the model simplifies to a single-dimensional diff.-in-disc. using credits passed as the running variable. This specification, shown below, isolates the effect of losing aid due to insufficient credit accumulation:

$$\begin{aligned}
Y_{it} = & \alpha + \beta(\widehat{Lose_{i,t+1} \times Recipient_{it}}) + \phi_1 Lose_{i,t+1} + \phi_2 Recipient_{it} + \\
& \phi_3 \tilde{R}_{it} + \phi_4(\tilde{R}_{it} \times Lose_{i,t+1}) + \phi_5(\tilde{R}_{it} \times Recipient_{it}) + \\
& \phi_6(\tilde{R}_{it} \times Lose_{i,t+1} \times Recipient_{it}) + X_i + W_{i,t-1} + \\
& Major_{i,t=1} + Institution_i + Cohort_i + \epsilon_{it}
\end{aligned} \tag{4}$$

where $\tilde{R}_{it}$ denotes centered credits passed and all other variables are defined as in Equation 3. The endogenous treatment indicator, $(Lose_{i,t+1} \times Recipient_{it})$, is instrumented using the interaction $(1\{Credits_{it} < 24\} \times Recipient_{it})$. The coefficient of interest, β , captures the LATE of losing the TEXAS Grant due to insufficient credit accumulation.

2. **GPA Discontinuity.** Represented by regions F, G, J, and K in Figure 3, this group includes students who are close to the GPA benchmark but satisfy the credit completion requirement. Students in this group passed at least 24 credits with cumulative GPAs ranging from 2.172 to 2.828 based on the optimal bandwidth around the 2.5 threshold. This group comprises 27.0% of the second-year sample and 22.0% of the third-year sample, and includes 56.3% and 30.8% of second- and third-year recipients, respectively, who lose the grant. Because all students in this group satisfy the credit requirement, this specification isolates the impact of aid loss due to narrowly missing the GPA threshold. The estimating equation is defined analogously to the Credit specification but uses centered GPA as the running variable and an indicator for falling below the 2.5 GPA threshold, interacted with recipient status, as the instrument.

¹¹Only students in region G are included in multiple specifications. Students in all other regions appear in only one group.

3. **Diagonal.** Students who narrowly meet or narrowly miss both renewal thresholds—those in regions C and G of Figure 3—comprise the third group. This includes students who are simultaneously within the optimal GPA and credit bandwidths and who fall either above both thresholds or below both thresholds. This group includes 14.5% of students in the second-year benchmark and 10.7% in the third-year benchmark. Similarly, it accounts for 18.7% and 15.8% of students who lose the TEXAS Grant at each respective benchmark. I retain the full model shown in Equation 3 but restrict the sample to students in regions C and G. This specification isolates the effect of aid loss among students who fall below both thresholds. Students in regions B and F are excluded because, although they are within the optimal bandwidths of both thresholds, they fall below only below one. Including them would conflate the effects of failing both benchmarks with the effects of failing just one.

Table 2 presents descriptive statistics for each subgroup, showing meaningful differences in both demographics and academic performance. The share of White students is highest in the Credit specification (46.6%), while the share of Black students is nearly double in the GPA specification compared to the Credit specification (16.0% vs. 8.5%). Further, students in the Credit specification had GPAs 0.68 points higher, on average, than those in the GPA specification, but they completed nearly 2 fewer credits. Those in the Diagonal group had the fewest completed credits (53.32) and the second-lowest GPAs (2.64) of the three groups. These patterns highlight the importance of estimating effects separately by region.

Variable	Full	Credit	GPA	Diagonal
Male	0.439	0.432	0.474	0.484
White	0.454	0.466	0.398	0.392
Black	0.100	0.085	0.160	0.146
Hispanic	0.326	0.322	0.354	0.374
Other	0.120	0.127	0.088	0.088
First-Generation	0.145	0.141	0.164	0.171
Credits Passed	55.01	54.55	56.34	53.32
Cumulative GPA	3.15	3.27	2.59	2.64
N	182,137	152,585	49,233	26,447

Notes: This table reports descriptive statistics for students in the second-year benchmark analysis. “Full” includes all students within the optimal bandwidth around at least one threshold. “Credit” includes students who met the GPA requirement but fell within the optimal bandwidth around the credit requirement. “GPA” includes students who met the credit requirement but fell within the optimal bandwidth around the GPA threshold. “Diagonal” includes students within the optimal bandwidth of both thresholds who were above or below both thresholds. Cumulative GPA and credits passed are measured at the end of the second year, when TEXAS Grant renewal is assessed.

Table 2. Descriptive Statistics by Sample

5.3 Identifying Assumptions

The validity of the diff.-in-disc. framework relies on satisfying two key assumptions: (1) continuity in observable characteristics across the threshold and (2) no manipulation of the running variables.

5.3.1 Covariate Balance

For the continuity in observable characteristics, any changes at the thresholds must be similar for both recipients and non-recipients. To test this assumption, I conduct a covariate balance check using the following equation:

$$\begin{aligned} X_{it} = & \alpha + \beta(1\{GPA_{it} < 2.5 \text{ or } Credits_{it} < 24\} \times Recipient_{it}) + \\ & \eta_1(1\{GPA_{it} < 2.5 \text{ or } Credits_{it} < 24\}) + \eta_2 Recipient_{it} + \\ & f(GPA_{it}, Credits_{it}) + [Recipient \times f(GPA_{it}, Credits_{it})] \end{aligned} \quad (5)$$

where X_{it} is a vector of student characteristics, previous academic outcomes (previously denoted as $W_{i,t-1}$ in Equation 3), and the institution attended. All other variables retain their previous definitions. In this equation, β captures whether there are systematic differences in student characteristics between recipients and non-recipients at the thresholds.

The three panels of Table 3 present results from the above equation for the second-year renewal benchmark. While most characteristics show no significant differences, a few small but notable shifts emerge. Moving from just above to just below a renewal threshold is associated with a 3.0 pp greater proportion of White students among recipients than non-recipients, and a 3.7 pp lower proportion of Hispanic/Latino students. Similarly, the share of first-generation students is 3.7 pp lower among recipients, and the share from families earning below \$20,000 is 1.6 pp higher.¹²

Panel C reports results for academic characteristics prior to renewal, which are especially important given that the outcomes of interest include credit accumulation, GPA, and graduation rates. It is important to verify that recipients and non-recipients are comparable on these pre-treatment academic measures across the thresholds; otherwise, any pre-existing differences could bias estimates. However, there is no evidence of differential changes in STEM representation, GPA, credits attempted or passed, or math and science credits attempted between recipients and non-recipients across the thresholds.¹³

Table A.2 shows that patterns in student demographics, family income, and institution attended for the third-year benchmark are similar to those in the second-year benchmark. However, small differences emerge in academic outcomes when falling below a renewal threshold. For recipients, GPA is 0.066 points higher and credits passed are 0.591 higher than for non-recipients.

Across both renewal benchmarks, there are some differences in student characteristics between

¹²These family income bins reflect the response categories on student applications.

¹³STEM majors are defined according to the Department of Homeland Security’s STEM Designated Degree Program list that includes majors under the following four primary CIP 2-digit series: 14 (Engineering), 26 (Biological and Biomedical Sciences), 27 (Mathematics and Statistics), and 40 (Physical Sciences) ([Department of Homeland Security, 2023](#)).

Panel A: Student Demographics		Panel B: Family Income		Panel C: 1st-Year Academic Outcomes	
Outcome	Estimate	Outcome	Estimate	Outcome	Estimate
Age	-0.002 (0.0141)	Unknown	-0.012 (0.0152)	STEM Major	0.020 (0.0139)
Male	-0.007 (0.0159)	<\$20,000	0.016*** (0.0077)	Cumulative GPA	0.001 (0.0080)
White	0.030** (0.0152)	\$20k–\$39,999	-0.015 (0.0094)	Credits Attempted	0.016 (0.1040)
Black	0.010 (0.0094)	\$40k–\$59,999	0.008 (0.0093)	Credits Passed	0.041 (0.1027)
Hispanic/Latino	-0.037** (0.0145)	\$60k–\$79,999	0.003 (0.0086)	Math Credits Attempted	0.089 (0.0964)
First-Generation	-0.037*** (0.0110)	\$80,000+	0.000 (0.0140)	Science Credits Attempted	0.198 (0.1526)
N	182,137	N	182,137	N	182,137

Notes: The family income bins in Panel B correspond to the options students can select on their applications. STEM majors include fields related to engineering, biological sciences, mathematics/statistics, and physical sciences. Mathematics and science courses are classified based on subject codes, with mathematics/statistics representing math courses and biology, chemistry, physics, physical sciences, geology, and general science representing science courses. *p<0.10; **p<0.05; ***p<0.01.

Table 3. Covariate Balance Check – Year 2 Benchmark

recipients and non-recipients across thresholds. However, these differences are generally small in magnitude, and overall patterns across thresholds are similar for both groups. As a result, they are unlikely to meaningfully bias the estimated effects of aid loss.

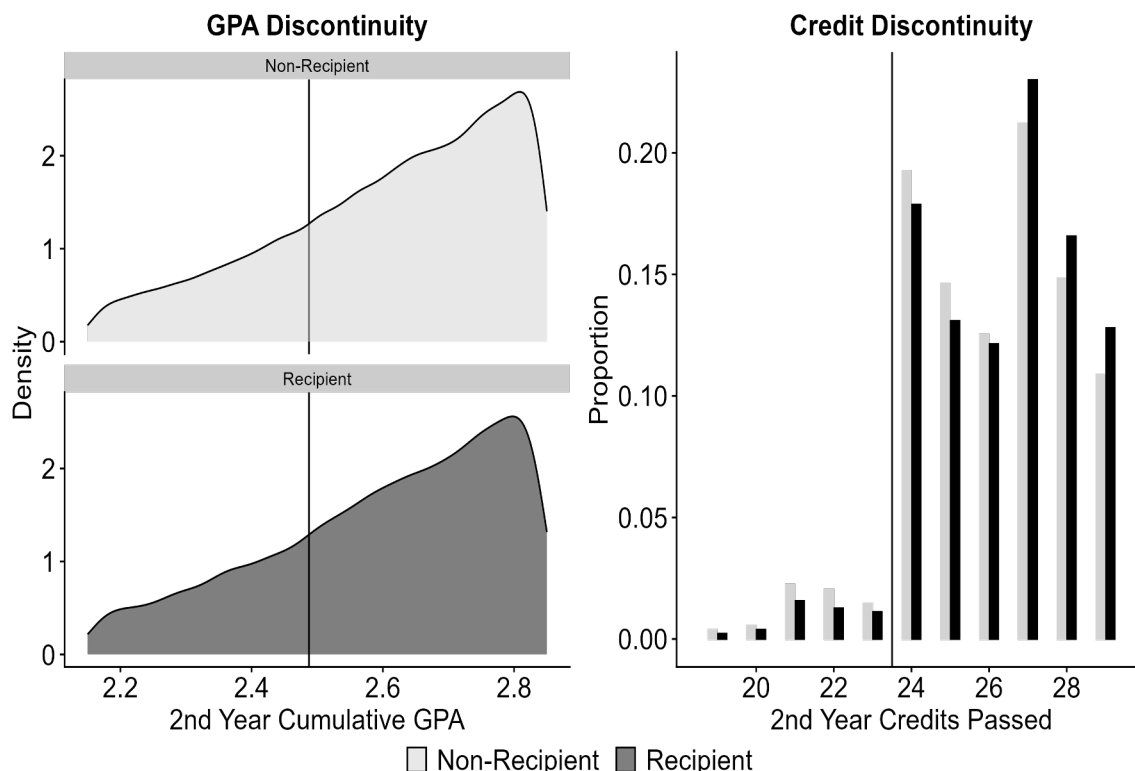
5.3.2 No Manipulation

Second, recipients must not be able to differentially manipulate either of the running variables. While some degree of sorting or strategic behavior may occur, what matters for identification is that any bunching or discontinuities at the threshold are similar for both recipients and non-recipients. If, for example, recipients disproportionately clustered just above the GPA renewal threshold while non-recipients were evenly distributed, this would suggest differential sorting that could bias estimates.

The two graphs in the left panel of Figure 4 plot the density of cumulative GPA for TEXAS Grant recipients and non-recipients at the second-year renewal benchmark. Visual inspection confirms that both groups exhibit similar density patterns around the threshold, suggesting no systematic manipulation of GPA. This aligns with the fact that students have limited ability to control their cumulative GPA. Moreover, unlike prior research documenting bunching at GPA values corresponding to standard letter grades (Barreca et al., 2016)—primarily in first-year GPA distributions due to grading scales and fewer courses contributing to GPA—there is no evidence of similar bunching at the second- and third-year renewal thresholds.

While students cannot precisely manipulate GPA, higher GPAs may still increase the likelihood

of grant retention among those who fall below a renewal threshold. As discussed earlier, Appendix Table A.1 shows that students who retain the grant despite not meeting the requirements tend to have significantly higher GPAs than those who lose it. This suggests that institutions may exercise some discretion in favor of stronger academic performers. However, this does not violate this identifying assumption: students cannot manipulate their GPA to ensure retention.



Notes: This figure shows the distribution of cumulative GPA (left) and credits passed (right) at the second-year renewal benchmark for TEXAS Grant recipients and non-recipients. The vertical lines indicate the relevant GPA (2.5) and credit (24) thresholds.

Figure 4. Density Plots – Year 2 Benchmark

The right panel of Figure 4 plots the distribution of students passing between 19 and 29 credits. If the renewal requirement were based on attempted credits, direct manipulation would be a major concern, as students could strategically adjust their course loads to stay above the threshold. However, just as students cannot directly manipulate their GPA, they also cannot directly manipulate whether they pass a course. While students can exert more effort throughout a course, this does not directly control whether a grade ultimately falls on the passing versus failing side of the margin. In this setting, a student would need either to perform sufficiently well late in the semester or to convince an instructor to change a failing grade to a passing grade.¹⁴

In theory, a student could attempt an extreme number of credits (e.g., 42) to increase the likeli-

¹⁴Appendix Figure A.2 plots the distribution of course letter grades and shows no excess mass for recipients at the lowest passing grade (D), consistent with the absence of fail-to-pass manipulation.

hood of passing at least 24, but this would be highly unusual given typical course loads and the risk of falling below a 2.5 GPA if the workload proves unmanageable.¹⁵ However, while there is visible bunching at and just above the 24-credit threshold, this simply reflects the fact that more students pass all of their courses than fail at least one. Crucially, the density shift across the threshold is nearly identical for recipients and non-recipients, indicating no differential manipulation. Thus, while there is some natural clustering around the cutoff, the similarity across groups supports the validity of the no manipulation assumption.

Figure A.3 presents density plots for the third-year renewal thresholds. Because only TEXAS Grant recipients were subject to a cumulative GPA requirement in the second year, their third-year GPA distribution is more concentrated above the 2.5 threshold, as shown in the left two graphs. However, there is no noticeable change in density across the threshold for either group, suggesting that differential manipulation of GPA remains unlikely.

The right-hand graph, in contrast, reveals a notable difference: a larger proportion of non-recipients passed 24 or more credits, while a slightly smaller share completed between 18 and 23 credits. This pattern suggests some degree of differential movement across the credit threshold, though it appears to be driven by non-recipients rather than recipients. While non-recipients lack a direct incentive to manipulate their credit accumulation in response to TEXAS Grant renewal rules, one possible explanation is the influence of institutional policies that apply to broader segments of the student population. Crucially, while this density shift points to differential movement across the threshold, it does not appear to reflect strategic manipulation among recipients in response to the TEXAS Grant renewal criteria.

6 Results

6.1 Second-Year Renewal Benchmark

6.1.1 Financial Aid

Table 4 presents the estimated impact of losing the TEXAS Grant after the second year on subsequent financial aid. Panel A reports effects on third-year financial aid, while Panel B shows effects on fourth-year financial aid for students who remained enrolled. Each column corresponds to one of the four specifications discussed in the previous section. The first column shows estimates from the Full specification, which includes all students near at least one renewal threshold. The remaining columns correspond to the three alternative specifications.

With the exception of students who narrowly missed both thresholds (Diagonal specification), losing the TEXAS Grant significantly reduced total third-year financial aid. The decrease in total aid ranged from \$3,429 in the Credit specification to \$4,509 for students in the GPA specification, representing an approximate 20% decrease in total aid. The primary driver of this decline was the

¹⁵Regressing credits attempted during the renewal year on a recipient indicator yields statistically insignificant estimates: recipients attempt 0.09 fewer credits in the second year and 0.26 fewer in the third year. This suggests that recipients are not strategically over-enrolling to create a buffer.

Aid Component	Specification				Specification			
	Full	Credit	GPA	Diagonal	Full	Credit	GPA	Diagonal
	Panel A: 3rd-Year Financial Aid				Panel B: 4th-Year Financial Aid			
Total Aid	-3574*** (456)	-3429*** (1020)	-4509*** (743)	-1503 (2027)	-1665*** (515)	-583 (1212)	-1844*** (826)	-674 (2324)
TEXAS Grant	-6183*** (87)	-5715*** (26)	-6415*** (142)	-5917*** (288)	-2631*** (170)	-2235*** (399)	-1468*** (283)	-2411*** (671)
Non-TG Gift Aid	963*** (276)	-368 (613)	807* (448)	855 (1130)	5 (300)	-450 (747)	-499 (469)	-1745 (1249)
Loans	1620*** (333)	2731*** (730)	978** (542)	3259** (1528)	939*** (361)	1922*** (805)	84 (582)	3385** (1646)
Federal Loan Maximum	0.093*** (0.026)	0.104* (0.056)	0.059 (0.042)	0.054 (0.107)	0.075*** (0.028)	0.213*** (0.067)	0.020 (0.045)	0.230* (0.122)
Parent PLUS	0.022 (0.022)	-0.096* (0.050)	0.002 (0.036)	0.135 (0.096)	0.007 (0.023)	-0.105** (0.050)	-0.020 (0.038)	0.075 (0.091)
Work-Study	25 (68)	-77 (165)	120 (111)	300 (258)	23 (66)	180 (164)	40 (105)	98 (256)
Earnings	18 (35)	-43 (97)	53 (50)	125 (165)	51 (42)	174* (89)	95 (64)	-287 (338)
Loans After Benchmark	–	–	–	–	1444 (1159)	5016* (2615)	-452 (1882)	9493* (4940)
Cumulative Federal Loan Maximum	–	–	–	–	0.027 (0.026)	0.000 (0.016)	0.019 (0.042)	0.128 (0.115)
N	182,137	152,585	49,233	26,447	157,982	133,220	42,373	23,479

Notes: 4th-year financial aid estimates are only provided for students who enrolled a fourth year. “Loans After Benchmark” is the cumulative loan amount students borrowed after their second year. The annual federal loan maximum for dependent students is \$7,500 in the third year and beyond. The cumulative federal loan maximum is \$31,000. *p<0.10; **p<0.05; ***p<0.01.

Table 4. Subsequent Financial Aid – Year 2 Benchmark

\$5,500–\$6,500 loss from the TEXAS Grant, although students partially offset this loss through other forms of aid. One such source is non-TEXAS Grant gift aid. Students in the Full and GPA specifications received just under \$1,000 more from these sources. To better understand the sources of this aid, Appendix Table A.3 presents estimates for the Full specification across all observed sources of third-year non-TEXAS Grant gift aid.

Several federal and state need-based aid programs exhibit modest increases following TEXAS Grant loss. Specifically, students received \$110 more from the Supplemental Educational Opportunity Grant (SEOG),¹⁶ \$330 more from HB3015,¹⁷ and \$201 more from the Texas Public Education Grant (TPEG).¹⁸ Taken together, these three programs account for approximately \$650 in additional aid. Relative to students’ typical second-year awards in these programs, this represents a large increase in need-based grant support, even though the total amount remains small compared to the TEXAS Grant award itself. These programs are all need-based and do not require separate applications, so the increases are consistent with awards adjusting upward when students have greater unmet need after losing the TEXAS Grant.

¹⁶The SEOG is a federal need-based grant administered by institutions. Awards are targeted toward Pell-eligible students with exceptional financial need and are subject to limited institutional allocations.

¹⁷HB3015 refers to state-authorized tuition-set-aside grants, which require institutions to set aside a portion of designated tuition revenue for need-based aid. Awards are typically reserved for Texas residents with financial need.

¹⁸TPEG is a state-administered, need-based grant available to Texas residents or students attending a public college in Texas. Eligibility is determined by financial need, and awards vary by institution and available funds.

In addition, students received \$1452 more in other forms of gift aid not shown separately in the table. This category captures smaller state, institutional, or private grants for which the specific source is not identified. While some of this increase may reflect the same mechanical response to higher financial need observed in other programs, it is also possible that students actively seek out new sources of grant aid after losing the TEXAS Grant. However, the data do not allow these mechanisms to be distinguished.

Merit-based aid displays a different pattern. Texas Top 10% scholarships increase slightly (\$25), while institutional merit aid declines by approximately \$600. This decline does not imply that institutional merit aid programs use the same renewal criteria as the TEXAS Grant. Rather, many merit scholarships impose academic standards that are stricter than the TEXAS Grant's 2.5 GPA requirement or are tied to broader measures of academic progress. As a result, students who fall below a TEXAS Grant renewal threshold may simultaneously lose certain merit awards, even though the eligibility rules are distinct. Taken together, these adjustments across other grant sources produce a net increase of just under \$1,000 in non-TEXAS Grant gift aid, though this replacement remains small relative to the TEXAS Grant award itself.

As increases in non-TEXAS Grant gift aid replace only a small share of the lost grant, students primarily turn to borrowing to smooth the financial shock. Loans represent the largest compensatory response to TEXAS Grant loss, though the magnitude of the increase varies across specifications. Third-year borrowing rises by roughly \$1,600 in the Full specification and \$2,700 in the Credit specification, while loan amounts rise by more than \$3,400 (80%) among students who fall below both renewal thresholds. This substantial borrowing response largely explains why total third-year aid does not decline significantly for this latter group. In contrast, students in the Full and Credit specifications are less able to fully replace lost grant aid through borrowing. In both groups, students are significantly more likely (9.3 pp and 10.4 pp, respectively) to borrow up to the annual federal loan maximum of \$7,500, despite only about 16% of aid recipients typically reaching this cap. This binding loan limit helps explain why total aid still falls sharply for these students even as borrowing increases.

To assess whether families compensate through alternative borrowing channels, I also examine Parent PLUS loans, which are federal loans taken out by a student's parent and are not subject to the same borrowing limits. These loans, however, require a credit check and are unavailable to families with adverse credit histories. Overall, I find little evidence that TEXAS Grant loss increases Parent PLUS borrowing. However, in the Credit specification, Parent PLUS take-up falls by nearly 9 pp, which may reflect reluctance or limited ability among families to take on additional parental debt.

Finally, I examine earnings from the federal work-study program and from outside employment. I find no significant changes in either source following TEXAS Grant loss, suggesting that students do not substantially increase labor supply in the short run to compensate for lost aid. In sum, students adjust to aid loss through a combination of increased borrowing and modest increases in other grant aid, but borrowing constraints, reluctance to take on additional debt, and potential

adjustment through credit channels not observed in the data limit their ability to fully offset the loss of TEXAS Grant funding, leading to meaningful reductions in total financial support during the third year.

By the fourth year, the effect of losing the TEXAS Grant on total aid diminished in magnitude, with reductions in total aid approximately half the size of the third-year estimates. This attenuation largely reflects a smaller direct effect of TEXAS Grant loss, as some students retained the grant through their third year but lost it afterward. That is, the third-year estimate captures the impact of losing aid after the second year relative to students who retained it into the third year. In contrast, the fourth-year estimate compares students who lost the grant after the second year to those who either retained it for four (or more) years or lost it after their third year.

Consistent with third-year trends, most students continued to rely on loans to partially offset the loss. In the Full specification, students borrowed an additional \$939, while those who fell below both thresholds continued to borrow at high levels, with a \$3,385 increase in fourth-year loans. Many also remained more likely to hit the annual federal loan maximum, with increases ranging from 7.5 to 23.0 pp. As in earlier years, there is no evidence that students increased their participation in work-study or significantly raised their outside earnings in response to the aid loss.

The final two rows assess the cumulative impact of TEXAS Grant loss on student borrowing after the second year. Students below only the credit benchmark borrowed \$5,016 more in federal loans over their remaining college years, while those below both benchmarks borrowed over \$9,000 more. Despite significant increases in third- and fourth-year loan uptake, the effect on total subsequent borrowing in the Full specification is not statistically significant. One possible explanation is that students who lost aid may have left college earlier, thereby reducing their need for additional loans. This is explored in later sections on persistence and degree completion.

The last row examines whether students were more likely to reach the cumulative federal loan limit of \$31,000 for dependent undergraduates. I find no evidence that TEXAS Grant loss increased the likelihood of hitting this cumulative borrowing cap, suggesting that while students increased annual borrowing in response to aid loss, most did not exhaust their lifetime federal loan eligibility.

6.1.2 Academic Outcomes

Table 5 presents estimated effects on subsequent academic outcomes. Panel A reports outcomes measured in the third year, while Panel B reports outcomes measured in the fourth year, conditional on persisting to the fourth year. Because persistence into the third year is a condition for observing aid loss, I do not estimate its effect on third-year persistence. I instead begin with outcomes measured at the start of that year, beginning with transfer rates. Across specifications, there is no meaningful effect of losing the TEXAS Grant on transfers to other in-state institutions.

The next several rows focus on enrollment patterns and academic performance. I find a substantial decline in enrollment intensity following aid loss. In the Full specification, full-time enrollment—defined as attempting at least 24 credits during the academic year—declined by 6.5 pp. This effect is considerably larger in the Credit and Diagonal specifications, at 23.0 pp and 28.1 pp, respectively.

Outcome	Specification				Specification			
	Full	Credit	GPA	Diagonal	Full	Credit	GPA	Diagonal
	Panel A: 3rd-Year Academic Outcomes				Panel B: 4th-Year Academic Outcomes			
Persistence	–	–	–	–	-0.032 (0.021)	-0.185*** (0.048)	0.000 (0.034)	-0.067 (0.096)
Transfer	0.000 (0.016)	-0.024 (0.042)	0.003 (0.022)	0.005 (0.064)	0.018 (0.016)	-0.002 (0.045)	0.025 (0.025)	-0.017 (0.077)
Full-Time Enrollment	-0.065** (0.032)	-0.230*** (0.079)	0.018 (0.050)	-0.281* (0.157)	0.009 (0.034)	-0.114 (0.076)	0.062 (0.054)	-0.152 (0.152)
Credits Attempted	-0.535 (0.522)	-3.793*** (1.263)	0.478 (0.833)	-3.425 (2.438)	0.008 (0.593)	-0.827 (1.382)	0.445 (0.940)	-2.055 (2.847)
Attempt 36+	0.008 (0.021)	-0.075 (0.046)	0.036 (0.035)	0.022 (0.078)	0.048** (0.021)	0.103** (0.047)	0.050 (0.035)	0.110 (0.099)
Credits Passed	-0.370 (0.594)	-3.909*** (1.431)	0.588 (0.947)	-3.473 (2.866)	0.110 (0.657)	-0.632 (1.524)	0.647 (1.047)	-1.589 (3.159)
GPA	0.046 (0.051)	0.088 (0.139)	0.120 (0.080)	-0.021 (0.257)	0.029 (0.060)	0.292 (0.155)	0.098 (0.096)	0.183 (0.276)
N	182,137	152,585	49,233	26,447	157,982	133,220	42,373	23,479

Notes: “GPA” only considers courses during the given year, not cumulative GPA. 4th-Year academic outcomes are conditional on students enrolling a fourth year. *p<0.10; **p<0.05; ***p<0.01.

Table 5. Subsequent Academic Outcomes – Year 2 Benchmark

These reductions in enrollment intensity are accompanied by sizable declines in credit accumulation among students below only the credit benchmark, who attempted 3.79 fewer credits and passed 3.91 fewer credits. In contrast, changes in credits attempted and passed are smaller and statistically insignificant in the Full and GPA specifications, and I find no evidence of changes in GPA across specifications.

Reduced enrollment intensity also has direct implications for students’ cost of attendance through lower tuition charges. However, these tuition reductions do not fully account for the decline in total aid observed in Table 4. Using the estimates from Table 5 and multiplying by the average tuition per credit among students who lost the TEXAS Grant, I estimate that tuition charges decreased by \$1,071 (3.793×282.43) in the Credit specification. After accounting for these reductions in tuition, total third-year aid remained \$2,076 ($-3147 + 1071$) lower. While reduced course loads modestly reduce tuition costs, they do not explain the full decline in aid, indicating that most of the observed reduction reflects a loss of funding rather than a lower cost of attendance.

Panel B of Table 5 presents results on fourth-year outcomes. I find continued evidence of negative effects on persistence for students below the credit benchmark, with persistence declining by 18.5 pp. In contrast, I find no statistically significant effect on fourth-year persistence in the Full specification. Given a baseline fourth-year persistence rate of 94.4% among third-year TEXAS Grant recipients, these estimates indicate that while most affected students remain enrolled, those who lost aid due to insufficient credit accumulation are substantially more likely to exit before the fourth year.

Among students who persisted, I find little evidence of continued declines in enrollment intensity or academic performance in the fourth year. Full-time enrollment and credits attempted and passed are not significantly affected across specifications, and I find no evidence of changes in GPA. Instead,

Outcome	Specification					
	Full			Credit		
	Third-Year FT Enrollment	Fourth-Year Persistence	Fourth-Year 36+ Credits	Third-Year FT Enrollment	Fourth-Year Persistence	Fourth-Year 36+ Credits
Recipient ($\beta_1 + \beta_3$)	-0.040*** (0.011)	-0.013 (0.009)	0.016*** (0.006)	-0.076*** (0.022)	-0.009 (0.017)	0.027*** (0.010)
Non-Recipient (β_1)	-0.011 (0.007)	0.001 (0.005)	-0.005 (0.003)	0.007 (0.014)	0.058*** (0.011)	-0.005 (0.007)
Difference (β_3)	-0.029** (0.013)	-0.014 (0.010)	0.021*** (0.007)	-0.083*** (0.026)	-0.067*** (0.020)	0.032** (0.012)
First-Stage (π_1)	0.438*** (0.005)	0.438*** (0.005)	0.438*** (0.005)	0.361*** (0.009)	0.361*** (0.009)	0.361*** (0.009)
Implied Effect (β_3/π_1)	-0.065	-0.032	0.048	-0.230	-0.185	0.088
N	182,137	182,137	157,982	152,585	152,585	133,220

Notes: This table decomposes selected estimates from Table 5 using pooled reduced-form and first-stage regressions. “Recipient” reports the implied discontinuity at the renewal threshold for TEXAS Grant recipients, while “Non-Recipient” reports the corresponding discontinuity for non-recipients. “Difference” reports the difference between these two discontinuities, capturing the reduced-form effect of aid loss. “First Stage” reports the pooled first-stage estimate for grant loss. “Implied Effect” reports the ratio of the reduced-form difference to the first-stage estimate and corresponds to the IV estimate reported in Table 5.

Table 6. Decomposing Effects of TEXAS Grant Loss

I observe some evidence of accelerated credit-taking. In the Full and Credit specifications, students were 4.8 pp and 10.3 pp more likely, respectively, to attempt 36 or more credits during their fourth year. These estimates suggest that some students may respond to earlier enrollment disruptions by increasing coursework in later years, potentially in an effort to remain on track for graduation. As in earlier years, I find no significant effects on academic outcomes for students below only the GPA threshold.

Table 6 uses pooled reduced-form and first-stage regressions to unpack selected estimates reported in Table 5. The pooled reduced form, shown in Equation 6, replaces the endogenous treatment indicator with an indicator for falling below a renewal threshold interacted with TEXAS Grant recipient status. From this specification, the table reports the implied discontinuity at the threshold for recipients ($\beta_1 + \beta_3$), the corresponding discontinuity for non-recipients (β_1), and their difference (β_3). The table then reports the pooled first-stage estimate (π_1 from Equation 1). Dividing the reduced-form difference by this first stage yields the implied effect of aid loss, which corresponds to the IV estimate reported in Table 5 and is shown in the final row of Table 6.

$$\begin{aligned}
Y_{i\tau} = & \alpha + \beta_1(1\{GPA_{it} < 2.5 \text{ or } Credits_{it} < 24\}) + \beta_2 Recipient_{it} + \\
& \beta_3[1\{GPA_{it} < 2.5 \text{ or } Credits_{it} < 24\} \times Recipient_{it}] + f(GPA_{it}, Credits_{it}) + \\
& [Recipient_{it} \times f(GPA_{it}, Credits_{it})] + X_i + W_{i,t-1} + \\
& Major_{i,t=1} + Institution_i + Cohort_i + \epsilon_{it}
\end{aligned} \tag{6}$$

Table 6 shows that the effects observed in the Full specification are driven almost entirely

by changes among TEXAS Grant recipients. For third-year full-time enrollment and fourth-year attempting 36+ credits, the reduced-form discontinuities for non-recipients are small and insignificant, while estimates for recipients are large and significant (-4.0 pp and 1.6 pp, respectively). A similar pattern holds in the Credit specification for most outcomes, with recipient discontinuities accounting for the majority of the reduced-form differences.¹⁹

However, fourth-year persistence is an exception. In this case, the reduced-form difference is driven by a positive discontinuity among non-recipients at the credit threshold, with no corresponding change for recipients. Consequently, the fourth-year persistence estimate in the Credit specification reflects enrollment behavior one year after the renewal benchmark rather than an immediate response to TEXAS Grant loss. By this point, the primary enrollment response to aid loss (third-year persistence) may have already occurred, and fourth-year persistence therefore captures later-stage enrollment dynamics. I explore third-year persistence in a later section, which provides additional context for interpreting this result.

Taken together, these results indicate that the consequences of TEXAS Grant loss depend critically on the criterion that triggers aid loss. Students who fell below only the credit threshold experienced the largest declines in enrollment intensity and academic progress, despite having relatively strong academic records prior to aid loss. At the same time, there is evidence of partial recovery in later years, as some increase course-taking intensity in the fourth year. In contrast, students who failed to meet only the GPA requirement exhibited little evidence of adverse academic effects. Students who failed to meet both requirements also display negative point estimates across several outcomes, though smaller sample sizes limit precision. These patterns highlight the importance of disaggregating effects by the specific renewal criterion, as estimates from the Full specification alone would mask substantial heterogeneity in how students respond to the loss of financial aid.

The findings underscore that financial aid loss does not exclusively affect academically struggling students. Instead, the largest academic disruptions are concentrated among students with high GPAs who lose aid due to insufficient credit accumulation. While the analysis cannot isolate mechanisms directly, one possibility is that aid loss alters students' perceptions of their academic standing or progress toward degree completion. In models of decision-making in college, students update beliefs about their ability and likelihood of success as they receive new information during college (Stinebrickner and Stinebrickner, 2012). In this context, the loss of financial aid, particularly when unexpected, may act as a salient negative signal, even for students who are otherwise performing well academically. The results therefore highlight how aid renewal rules can generate meaningful academic consequences through financial constraints and perceived academic risk, rather than academic ability alone.

¹⁹Appendix A.4 plots binned means and local linear fits around the credit renewal threshold for the outcomes in Table 6, separately for recipients and non-recipients.

6.1.3 Graduation

Table 7 presents estimates on four-, five-, and six-year graduation rates, degree attainment, and time to degree.²⁰ Overall, there is little evidence that losing the TEXAS Grant affects degree attainment or graduation timing across most specifications and horizons. The one exception is a positive effect (5.9 pp) on four-year graduation in the Full specification. This result is consistent with earlier evidence that some students increase course-taking intensity in the fourth year, potentially reflecting efforts to accelerate progress toward degree completion and limit future borrowing. I find no statistically significant effects on five- or six-year graduation rates, degree attainment, or time to degree in other specifications.

Outcome	Specification			
	Full	Credit	GPA	Diagonal
Graduate in 4 Years	0.059* (0.031)	0.078 (0.072)	0.021 (0.052)	-0.154 (0.110)
N	182,137	152,585	49,233	26,447
Graduate in 5 Years	0.043 (0.034)	0.015 (0.084)	0.006 (0.054)	-0.167 (0.158)
N	159,618	132,634	44,933	24,049
Graduate in 6 Years	0.029 (0.034)	-0.016 (0.087)	0.055 (0.054)	-0.292* (0.175)
Bachelor's Degree	-0.011 (0.031)	-0.097 (0.079)	0.025 (0.049)	-0.195 (0.167)
N	135,531	112,027	39,076	20,802
Time to Graduate (Months)	-1.498 (1.036)	-4.417 (2.889)	-0.369 (1.614)	3.935 (5.211)
N	152,404	129,546	39,084	20,546

Notes: Four-year graduation rates include all cohorts; five-year graduation rates omit the fall 2018 entry cohort as spring 2023 graduation records are not yet available; six-year graduation rates and degree attainment further omit the the fall 2017 entry cohort. Time to graduation only includes graduates, using the month and year of their graduation and assumes a uniform starting month of September for all universities. *p<0.10; **p<0.05; ***p<0.01.

Table 7. Graduation Outcomes – Year 2 Benchmark

6.1.4 Major Change

Financial aid loss can create strong incentives for students to reevaluate their major selection, but the likelihood of switching depends on several competing factors. On one hand, the loss of financial support may increase pressure to reduce academic demands or costs, encouraging students to consider changing majors. On the other hand, switching can itself be costly, especially later in

²⁰Table A.4 shows that the results are similar when restricting to a fixed set of cohorts across outcomes, confirming that the main findings are not driven by which cohorts are included.

college. Many majors have distinct course sequences and requirements, so changing fields may delay graduation and increase overall educational expenses—a risk that financially constrained students may be especially reluctant to take. The scope of the major change may also matter: switching to a substantially different field (e.g., from Engineering to Psychology) would require additional prerequisite coursework and extend time to degree.

Among those who do switch majors, there are likewise competing forces that may shape the nature of the change. The increased debt burden resulting from aid loss could incentivize students to pursue higher-earning fields in anticipation of future repayment obligations (Rothstein and Rouse, 2011). Another possibility is that students increase work hours in response to financial strain. While I do not find significant effects on reported earnings, prior research finds that financial aid loss can increase work hours (Carruthers and Özek, 2016), which could in turn push students toward less demanding majors. Similarly, a desire to graduate more quickly may also drive students to switch into more flexible or closely related majors that facilitate faster credit accumulation. These diverging responses highlight the complex ways financial instability can reshape academic decision-making.

To explore these possible mechanisms, Table A.5 presents estimated impacts on the likelihood of a student changing majors in the year following the loss of the TEXAS Grant, two years after aid loss (conditional on remaining enrolled), and by graduation (conditional on graduating). For students who changed majors, the table also reports estimates on whether the change involved moving across 2-digit CIP families or entering or exiting a STEM field.²¹

Table A.5 shows no evidence that losing the TEXAS Grant changes the likelihood of switching majors. Across specifications, estimated effects on major switching after one year, after two years, and by graduation are small and statistically insignificant. Among those who did switch, however, I observe some substantive changes in field of study.

To assess the nature of these academic shifts, I first examine whether students who change majors move into a different two-digit CIP category, which captures more substantial changes in academic focus. Here, effects are concentrated in the Credit specification: students who switch majors are 19.7 pp less likely to exit their original CIP family after one year, but this effect does not continue after two years or at graduation. This pattern is consistent with the notion that large academic pivots may be costly.

I next examine movement into and out of STEM fields among students who switch majors. Here, the evidence points to short-run avoidance of STEM fields following aid loss. Across several specifications, students who change majors are less likely to move into STEM and, in some cases, more likely to move out of STEM in the years immediately following aid loss. These effects are not sustained by graduation, suggesting that while aid loss may temporarily discourage movement into more demanding or time-intensive fields, it does not lead to differences in degree field at completion.

Overall, these results indicate that aid loss does not increase the likelihood that students change majors. Instead, students who lose aid are less likely to make large or disruptive changes in academic

²¹STEM majors are defined based on CIP classifications and include engineering, biological sciences, mathematics/statistics, and physical sciences.

focus following renewal. Among the subset of students who do switch majors, there is evidence of short-run differences in the types of switches undertaken, but these differences do not persist to graduation.

6.1.5 Early-Career Earnings

This section shifts focus to longer-term outcomes by examining the early-career earnings effects of financial aid loss. These outcomes reflect the cumulative consequences of disrupted academic progress, including degree completion and major choice. While earlier results show no effects in students' final majors, career trajectories can still diverge substantially even within the same field of study. For example, students may pursue different types of jobs, industries, or locations depending on financial constraints or debt burdens. As a result, observed effects on earnings represent the net outcome of these competing forces.

To disentangle the combined effects of academic disruption and post-college choices, Table A.6 presents estimated effects on earnings one, two, and three years after college exit, reported separately for all students and for graduates.²² Across specifications, I find no statistically significant effects of losing the TEXAS Grant on post-college earnings at any horizon. Estimates are generally small and imprecise for both all students and graduates, and I find no consistent patterns by specification or time since exit. Thus, despite meaningful effects on enrollment intensity and borrowing during college, financial aid loss does not translate into detectable differences in early-career earnings among students observed in the Texas workforce data.

6.2 Third-Year Renewal Benchmark

This section examines whether similar effects emerge for students who face aid loss after the third year. Many of the same mechanisms likely contribute, but students reaching this checkpoint have already persisted further in college, signaling greater academic commitment and leaving them with fewer opportunities to adjust academic plans without delaying graduation.

6.2.1 Financial Aid

Table 8 presents estimates of the effects of losing the TEXAS Grant at the third-year renewal checkpoint on fourth-year financial aid. Overall declines in total aid are broadly similar to those observed following second-year aid loss. Most aid sources also follow similar patterns: students partially replaced lost aid with other grants and/or increased borrowing, with no significant change in work-study or outside earnings. Students below only the credit threshold were again more likely to reach the annual federal loan maximum (22.1 pp) and borrow significantly more after the third-year benchmark (\$5,749). In addition, these students were 19.4 pp more likely to reach the cumulative

²²This analysis captures earnings for approximately 63% of students. It includes only those who exited college and matched to TWC records; students still enrolled, in graduate school, unemployed in Texas, or working out of state are not included.

federal loan maximum despite having received the TEXAS Grant for three years, highlighting the extent to which aid loss can reshape borrowing trajectories deep into college.

Aid Component	Full	Credit	GPA	Diagonal
Total Aid	-3582*** (967)	699 (1844)	-5049*** (1503)	-12378 (7689)
TEXAS Grant	-5957*** (169)	-4989*** (320)	-6188*** (288)	-6449*** (1000)
Non-TG Gift Aid	1230** (618)	1292 (1147)	970 (926)	-2706 (3947)
Loans	1291* (724)	4450*** (1333)	174 (1148)	-2487 (5165)
Federal Loan Maximum	0.091 (0.061)	0.221** (0.109)	-0.024 (0.097)	-0.022 (0.378)
Parent PLUS	0.025 (0.046)	0.037 (0.088)	0.030 (0.071)	0.076 (0.341)
Work-Study	-145 (145)	-53 (262)	-5 (218)	-735 (1269)
Earnings	-14 (88)	-224 (182)	106 (131)	173 (292)
Loans After Renewal	432 (1939)	5749* (3300)	-1590 (3246)	-6487 (12825)
Cumulative Federal Loan Maximum	0.025 (0.058)	0.194* (0.105)	-0.057 (0.093)	-0.520 (0.425)
N	115,399	99,984	25,422	12,291

Notes: “Loans After Benchmark” is the cumulative loan amount students take out after their third year. The annual federal loan maximum for dependent students is \$7,500 in the third year and beyond. The cumulative federal loan maximum is \$31,000. *p<0.10; **p<0.05; ***p<0.01.

Table 8. Subsequent Financial Aid – Year 3 Benchmark

6.2.2 Academic Outcomes

Table 9 presents estimates on fourth-year academic outcomes. As in earlier results, the largest effects are concentrated among students who lost aid due to insufficient credit accumulation. In the Credit specification, aid loss is associated with a 9.7 pp increase in transfers to other institutions, indicating that some students respond to late-stage aid loss by leaving their original institution. Transfer effects are smaller in the Full specification (3.6 pp) and not significant in the GPA specification.²³

In addition to increased transfers, students in the Credit specification again experienced meaningful declines in credit accumulation. Credits attempted declined by 3.42 credits and credits passed fell by 3.58 credits, with no corresponding change in fourth-year GPA. Effects in the Full and GPA specifications are smaller and not statistically significant. As before, estimates for students failing both benchmarks are large but imprecise due to smaller sample sizes.

²³These transfer responses are driven by TEXAS Grant recipients and are not characterized by moves to lower-cost institutions or institutions closer to home, suggesting that they reflect broader adjustments to enrollment trajectories following aid loss rather than simple cost minimization.

Outcome	Full	Credit	GPA	Diagonal
Transfer	0.036* (0.021)	0.097* (0.050)	0.011 (0.027)	0.367* (0.191)
Full-Time Enrollment	-0.040 (0.066)	-0.093 (0.131)	0.053 (0.098)	-0.510 (0.498)
Credits Attempted	-1.317 (1.080)	-3.420*** (2.111)	0.233 (1.621)	-14.208 (9.117)
Attempt 36+	-0.014 (0.040)	-0.014 (0.073)	0.038 (0.064)	-0.257 (0.253)
Credits Passed	-1.465 (1.206)	-3.584 (2.431)	0.638 (1.764)	-11.841 (9.847)
GPA	-0.144 (0.111)	0.007 (0.255)	-0.003 (0.158)	0.155 (0.894)
N	115,399	99,984	25,422	12,291

Notes: “GPA” only considers courses during the given year, not cumulative GPA. *p<0.10; **p<0.05; ***p<0.01.

Table 9. Subsequent Academic Outcomes – Year 3 Benchmark

6.2.3 Graduation

Table 10 presents effects on graduation outcomes.²⁴ Across specifications, there is no statistically significant effects of aid loss on any graduation outcome. Unlike the earlier benchmark, losing aid at this later stage does not appear to meaningfully alter students’ graduation pathway.

The absence of detectable graduation effects at this stage is not surprising. Students who lose aid after three years of enrollment are already close to degree completion and may have fewer remaining requirements, making exit less likely even in the face of financial disruption. As a result, while earlier analyses show that late-stage aid loss affects enrollment intensity and increases transfers, these adjustments do not translate into measurable differences in graduation outcomes.

6.2.4 Major Change

Appendix Table A.8 presents estimated effects of aid loss on major switching. In contrast to the second-year benchmark, there is evidence that losing the TEXAS Grant at this later stage increased the likelihood of changing majors. In the Credit specification, students were 21.0 pp more likely to switch majors within one year of aid loss, indicating a sizable response along this margin. Among students who ultimately graduated, switching increased by 17.3 ppp in the Full specification, 29.7 pp in the Credit specification, and 26.3 pp in the GPA specification. GPA-based aid loss also reduced the likelihood of switching out of a STEM field by 19.0 pp. Beyond this result, I find no consistent evidence of directional movement into or out of STEM fields or systematic transitions across CIP families among students near any renewal threshold. This pattern suggests that financial aid loss can have disruptive effects on academic decisions even at very late stages of students’ college careers.

²⁴Table A.7 shows that the results are similar when restricting to a fixed set of cohorts across outcomes, confirming that the main findings are not driven by which cohorts are included.

Outcome	Full	Credit	GPA	Diagonal
Graduate in 4 Years	-0.058 (0.063)	0.062 (0.123)	-0.092 (0.106)	-0.379 (0.313)
N	115,399	99,984	25,422	12,291
Graduate in 5 Years	-0.103 (0.071)	-0.132 (0.150)	-0.011 (0.105)	-0.290 (0.415)
N	100,000	85,782	23,437	11,253
Graduate in 6 Years	-0.068 (0.064)	-0.076 (0.128)	-0.012 (0.097)	-0.208 (0.392)
Bachelor's Degree	-0.012 (0.056)	-0.121 (0.112)	-0.056 (0.085)	-0.336 (0.361)
N	85,435	72,775	20,770	9,922
Time to Graduate (Months)	2.488 (2.125)	2.514 (4.197)	1.077 (3.305)	0.751 (11.190)
N	103,528	90,247	21,953	10,112

Notes: Four-year graduation rates include all cohorts; five-year graduation rates omit the fall 2018 entry cohort as spring 2023 graduation records are not yet available; six-year graduation rates and degree attainment further omit the fall 2017 entry cohort. Time to graduation only includes graduates, using the month and year of their graduation and assumes a uniform starting month of September for all universities. *p<0.10; **p<0.05; ***p<0.01.

Table 10. Graduation Outcomes – Year 3 Benchmark

6.2.5 Early-Career Earnings

Appendix Table A.9 reports the estimated impact of losing aid after the third-year renewal checkpoint on early-career earnings. I find no evidence of earnings effects in the Full or GPA specifications at any post-exit horizon. In contrast, the Credit specification shows large and statistically significant increases in earnings in the first two years after exit, with estimates that decline in later years. A plausible interpretation is that these students select into jobs with higher initial earnings in response to higher student debt. The pattern over time is consistent with higher initial earnings accompanied by flatter subsequent earnings growth, rather than persistent differences in earnings trajectories.

Taken together, these results indicate that losing aid has meaningful effects on students' academic trajectories, with impacts concentrated along margins related to short-run academic progress. Effects are largest for students who lose aid due to insufficient credit accumulation, who experience large increases in loan borrowing and reductions in credit accumulation. However, these effects are largely temporary, as impacts on graduation, major choice, and early-career earnings are limited. The following section examines persistence and graduation outcomes for students who fall below a renewal threshold, removing the condition that students persist into the subsequent academic year.

6.3 Reduced-Form Estimates

This section relaxes the requirement that students persist into the subsequent academic year in order to examine outcomes for all students who fall below a renewal threshold. Students learn about their TEXAS Grant renewal status before the start of the next academic year, and some may respond by not enrolling at all. Conditioning on subsequent enrollment therefore excludes an important margin of adjustment and may understate the broader consequences of failing to meet renewal requirements. By focusing on the reduced-form effects of falling below a threshold—rather than aid loss itself—this analysis captures responses to renewal ineligibility that occur prior to observed aid loss.

Outcome	Second-Year Benchmark				Third-Year Benchmark			
	Specification				Specification			
	Full	Credit	GPA	Diagonal	Full	Credit	GPA	Diagonal
Third-Year Persistence	-0.002 (0.007)	-0.078*** (0.007)	0.002 (0.009)	0.003 (0.037)	—	—	—	—
Fourth-Year Persistence	-0.015 (0.011)	-0.142*** (0.011)	0.002 (0.016)	-0.039 (0.065)	-0.019 (0.017)	-0.177*** (-0.013)	-0.015 (0.021)	-0.169* (0.087)
N	193,007	161,855	51,534	27,556	129,510	112,650	27,680	13,230
Bachelor's Degree	0.003 (0.011)	-0.052*** (0.011)	0.011 (0.017)	-0.133* (0.079)	-0.008 (0.013)	-0.013 (0.010)	-0.023 (0.024)	-0.299** (0.122)
N	144,251	119,395	41,022	21,688	96,232	82,316	22,723	10,727

Notes: Estimates report reduced form effects of falling below a renewal threshold, not conditional on persisting to a third (fourth) year. *p<0.10; **p<0.05; ***p<0.01.

Table 11. Persistence and Graduation Outcomes – Reduced Form

Table 11 presents reduced-form estimates of falling below a renewal threshold on persistence and degree attainment at the second- and third-year benchmarks. The most pronounced effects appear at the initial post-renewal enrollment margin. For the second-year benchmark, falling below a threshold reduces third-year persistence by 7.8 pp in the Credit specification, and the effect is even larger for fourth-year persistence (14.2 pp). There are no significant effects in the other specifications. For the third-year benchmark, the analogous initial decision—persistence into the fourth year—declines by 17.7 pp in the Credit specification and 16.9 pp in the Diagonal specification, again with insignificant effects elsewhere. Turning to degree attainment, there are statistically significant declines in bachelor's degree completion in some specifications, including a 5.2 pp reduction in the second-year Credit specification and sizable negative effects for the Diagonal specification at both benchmarks (13-30 pp).

Table 12 decomposes the reduced-form estimates for students below only the credit threshold by recipient status. The decomposition shows consistently negative and statistically significant effects across outcomes for TEXAS Grant recipients, indicating that falling below the renewal threshold

Outcome	Second-Year Benchmark			Third-Year Benchmark	
	Third-Year Persistence	Fourth-Year Persistence	Bachelor's Degree	Fourth-Year Persistence	Bachelor's Degree
Recipient ($\beta_1 + \beta_3$)	-0.040*** (0.008)	-0.052*** (0.013)	-0.093*** (0.013)	-0.071*** (0.015)	-0.049*** (0.012)
Non-Recipient (β_1)	0.038*** (0.007)	0.090*** (0.010)	-0.041*** (0.010)	0.106*** (0.013)	-0.037*** (0.010)
Difference (β_3)	-0.078*** (0.010)	-0.142*** (0.017)	-0.052*** (0.016)	-0.177*** (0.020)	-0.013 (0.016)
N	161,855	161,855	119,395	112,650	82,316

Notes: This table decomposes the Credit specification estimates from Table 11. “Recipient” reports the implied discontinuity at the renewal threshold for TEXAS Grant recipients, while “Non-Recipient” reports the corresponding discontinuity for non-recipients. “Difference” reports the difference between these two discontinuities, capturing the reduced-form effect of falling below the credit threshold and corresponds to the estimate reported in Table 11.

Table 12. Decomposing Effects of Falling Below Credit Threshold

and facing the loss of eligibility for continued aid is associated with substantial reductions in persistence and degree attainment.

Non-recipients exhibit a different pattern. Persistence increases at the threshold for non-recipients, which may reflect fewer opportunities to transfer to out-of-state or private institutions or determination to recover and remain enrolled.²⁵ Prior work shows that higher-achieving students who are attentive to indicators of potential failure and perceive outcomes as controllable often respond by persisting rather than withdrawing, even following negative academic feedback (Perry et al., 2005).

These dynamics also help explain the selection underlying the earlier estimates of fourth-year persistence in Table 6. Because those estimates condition on remaining enrolled into the third year, they are estimated on a selected group of students. Among non-recipients, the increase in third-year persistence means that those who remain enrolled below the threshold are a more persistent and attached group, which in turn leads to higher fourth-year persistence. Recipients, by contrast, adjust primarily at the third-year margin, leaving little scope for subsequent persistence responses.

This pattern is also relevant for interpreting the IV estimates of aid loss on graduation as they too condition on an enrollment margin that is itself affected by falling below the renewal threshold. Since falling below the renewal threshold reduces third-year persistence among TEXAS Grant recipients, early exits following aid ineligibility are excluded from the aid-loss analysis. As a result, the graduation effects estimated earlier may understate the full impact of aid loss by abstracting from early exits that occur before aid loss is realized.

These reduced-form results highlight that renewal thresholds affect student outcomes not only through realized aid loss, but also through enrollment decisions at the point of ineligibility. Persistence into the subsequent year is itself an important margin of adjustment, and conditioning on this margin shapes the interpretation of later outcomes.

²⁵Persistence is measured within the Texas public higher education system; transfers to private or out-of-state institutions are not observed.

6.4 Discussion

Taken together, the findings throughout this section indicate that losing the TEXAS Grant primarily disrupts students' enrollment behavior and academic progress rather than their academic performance. The most salient responses occur along margins related to enrollment intensity, credit accumulation, and persistence, indicating that aid loss alters how students engage with college rather than how they perform in coursework.

The reduced-form analysis complements these findings by showing that renewal thresholds affect outcomes even before aid loss is realized. Falling below a renewal threshold induces changes in persistence and enrollment decisions at the point of ineligibility, revealing an important margin of adjustment that is not captured when conditioning on continued enrollment. These early responses help clarify how later persistence and graduation outcomes should be interpreted and underscore that renewal-based policies shape both the timing and nature of student exit.

Across both sets of results, the effects of renewal-based aid policies vary systematically by timing and by the criterion that triggers ineligibility. Responses are more pronounced around the second-year renewal checkpoint than later in students' academic careers, and students who fall short of only credit requirements—despite maintaining solid GPAs—exhibit the most severe disruptions. Together, these patterns suggest that credit-based renewal rules can be especially consequential for students who remain academically capable but experience temporary setbacks in credit accumulation. The next section examines how these effects differ across student and institutional contexts.

7 Heterogeneity

To assess how the impacts of losing the TEXAS Grant differ across institutional settings and student groups, I examine heterogeneity by institutional characteristics (tuition caps) and student background (race/ethnicity and first-generation status). These analyses focus on selected academic outcomes measured in the third year and degree attainment outcomes, and are restricted to the second-year benchmark and the Full and Credit specifications.

7.1 No Tuition Cap vs. Tuition Cap

To understand how institutional pricing structures influenced the consequences of losing financial aid, I examine heterogeneity by tuition cap status. Since the 2005 statutory reforms, all public universities in Texas charge a \$50 per-credit statutory tuition rate, as mandated by Texas Education Code §54.051(c) ([Texas Legislature, 2023](#)). However, designated tuition—the institutional component layered on top of statutory tuition—varies across institutions. Some universities charge a flat per-credit rate regardless of total enrollment, while others implement a cap (typically at 12 credits), beyond which additional credits incur no further designated tuition. Approximately one-third of public universities follow this capped model, representing roughly 30% of the students in my analysis.

This institutional variation shapes the marginal cost of enrolling in heavier course loads. At universities with a tuition cap, students pay only the \$50 statutory tuition for each credit beyond 12. In contrast, students at uncapped institutions continue to pay both the \$50 statutory rate and the institution’s designated tuition, making each additional course substantially more expensive. These pricing differences may influence students’ responses to aid loss, particularly when deciding whether to maintain or accelerate their academic progress.

For example, a student enrolled in 15 credits would pay just \$150 more than a 12-credit student at a capped institution, since the additional three credits would incur only the \$50 statutory tuition per credit. At an uncapped institution, however, each additional credit would cost both the \$50 statutory tuition and approximately \$200 in designated tuition, or \$750 for three credits.²⁶ While this gap may appear modest over one semester, it compounds over time: enrolling in three extra credits each semester for two academic years would cost an additional \$2,400 at an institution without a cap.

Crucially, these marginal cost differences only come into play after students lose the TEXAS Grant, which previously covered full tuition and fees. Once students must pay out of pocket, the financial burden of taking more than 12 credits becomes much more salient—especially at uncapped institutions. This may discourage heavier enrollment, increase loan reliance, or even prompt students to leave college due to rising costs. Although overall tuition levels vary, this example illustrates how pricing structures can amplify the academic consequences of aid loss and shape students’ post-renewal decisions.

The academic outcomes in Panel A of Table 13 show that responses to aid loss differ meaningfully across institution type. In the Full specification, aid loss is associated with an 8.3 pp decline in full-time enrollment at uncapped institutions, while the corresponding estimate at capped institutions is an insignificant 2.5 pp. Similarly, among students below only the credit threshold, full-time enrollment declines only occur at uncapped institutions (28.6 pp). For attempted credits, differences by cap status are concentrated in the Credit specification, where there is no significant effect at capped institutions, while students at uncapped institutions reduce attempted credits by 4.56. Results for attempting 36 or more credits in the third year show no significant effects at either capped or uncapped institutions, suggesting that aid loss does not meaningfully change the likelihood that students attempt especially heavy course loads in the third year.

Graduation outcomes in Panel B of Table 13 suggest more limited differences across institution types. In the Full specification, aid loss has no statistically significant effect on four-year graduation rates, degree attainment, or time to degree at either capped or uncapped institutions. However, among students at capped institutions, aid loss increased four-year graduation rates by 12.2 pp, with no comparable effect appearing at uncapped institutions. This pattern is consistent with students being able to increase course-taking intensity in ways that facilitate earlier completion without incurring higher tuition charges. Importantly, this result does not contradict the absence of

²⁶The \$200 designated tuition rate reflects an approximate average across uncapped Texas public universities; actual rates vary by institution and year.

	No Tuition Cap		Tuition Cap	
Specification	Full	Credit	Full	Credit
Panel A: 3rd-Year Academic Outcomes				
Full-Time Enrollment	-0.083** (0.040)	-0.286*** (0.093)	-0.025 (0.053)	-0.056 (0.151)
Credits Attempted	-0.512 (0.648)	-4.564*** (1.482)	-0.548 (0.878)	-1.289 (2.531)
Attempt 36+ Credits	0.021 (0.026)	-0.068 (0.054)	-0.020 (0.037)	-0.090 (0.095)
N	127,824	105,755	54,313	46,830
Panel B: Graduation Outcomes				
Graduate in 4 Years	0.039 (0.038)	0.041 (0.084)	0.122** (0.053)	0.191 (0.151)
N	127,824	105,755	54,313	46,830
Bachelor's Degree	-0.011 (0.038)	-0.125 (0.093)	-0.004 (0.052)	-0.017 (0.153)
N	95,622	77,991	39,909	34,036
Time to Graduate (Months)	-1.246 (1.297)	-3.421 (3.585)	-2.504 (1.703)	-7.017 (4.582)
N	105,613	88,485	46,791	41,061

Notes: Four-year graduation rates include all cohorts; degree attainment further omit the fall 2017 and 2018 entry cohorts. Time to graduation only includes graduates, using the month and year of their graduation and assumes a uniform starting month of September for all universities. *p<0.10; **p<0.05; ***p<0.01.

Table 13. Academic and Graduation Outcomes, By University Tuition Caps – Year 2 Benchmark

significant effects on attempting 36 or more credits in the third year, as that measure captures only one margin of enrollment intensity and only in a single year. Despite evidence of faster completion among some students at capped institutions, aid loss does not increase overall degree attainment, indicating that these timing responses do not translate into higher rates of eventual graduation. Taken together, the academic and graduation results suggest that tuition cap status is associated with meaningful differences in how students adjust enrollment behavior following aid loss, consistent with higher marginal costs of credit accumulation at uncapped institutions limiting students' ability to maintain or accelerate progress toward timely completion.

7.1.1 Student Characteristics

To further explore heterogeneity in the effects of TEXAS Grant loss, I explore heterogeneity by race/ethnicity and first-generation status. Racial/ethnic differences are motivated by persistent racial wealth gaps and disparities in access to need-based aid, which may amplify the consequences of aid loss (Sullivan et al., 2015; Espinosa et al., 2019). Likewise, first-generation students may be especially vulnerable to aid disruptions due to limited access to institutional knowledge and support systems (Pascarella et al., 2004; Unverferth et al., 2012; Ko et al., 2025).

Table A.10 reports results separately for Hispanic/Latino and White students. The most pronounced differences appear in short-term academic outcomes in the Credit specification. Hispanic/Latino students experience large and statistically significant disruptions following aid loss, including a 31.4 ppp decline in full-time enrollment and a reduction of 3.77 attempted credits. In contrast, for White students, estimates for both outcomes are smaller in magnitude and not significant. Graduation outcomes are more similar across racial/ethnic groups. There are no significant effects on four-year graduation rates or overall degree attainment for either group in either specification. However, there is some evidence of decreased time to graduation among White students, suggesting modest differences in degree timing rather than completion.

Table A.11 examines whether the effects of aid loss differ by first-generation status.²⁷ First-generation students experience substantial short-run academic disruptions following aid loss. In the third year, they exhibit large declines in full-time enrollment, credits attempted, and the likelihood of attempting 36 or more credits, with most estimates significant in both the Full and Credit specifications. In contrast, non-first-generation students experience smaller effects that are generally not statistically significant.

Differences in graduation outcomes are also pronounced. In the Credit specification, non-first-generation students are more likely to graduate within four years and complete their degrees more quickly, with time to degree falling by over a year. By contrast, first-generation students are 29.1 pp less likely to graduate within four years and 33.1 pp less likely to earn a bachelor's degree. Taken together, these results suggest that aid loss imposes particularly severe and persistent consequences for first-generation students, affecting both short-run academic progress and longer-run degree completion.

7.1.2 Course Failure Level and Type

I also explored whether the academic context of a failed course moderates the effects of aid loss. For example, failing a lower-level general education course may carry different implications than failing an upper-level course within a student's major. While I cannot directly observe whether a course fulfills a major requirement, I categorize failures by course level (100/200 vs. 300/400).²⁸ The estimated effects of aid loss are similar across these groups. This suggests that while course level may be academically meaningful, it is less relevant to the financial and behavioral consequences of losing aid than student-level characteristics, such as first-generation status. I also compared outcomes for students receiving incompletes (5% of failures) versus failing grades (Fs or No Credit) and found no meaningful differences. Excluding students with incompletes does not materially affect any of the main results presented in this paper. This suggests that course-level distinctions may play a

²⁷First-generation status is derived from parent education records. Students are classified as first-generation if neither parent attended college, and as non-first-generation if at least one parent did. Students are excluded if one parent's education is unknown and the other did not attend college, or if both are missing. As a result, the combined sample size is smaller than in the full analysis.

²⁸Among students who failed a course in their second year, roughly two-thirds failed a 100- or 200-level course; by the third year, roughly two-thirds failed a 300- or 400-level course.

smaller role in shaping the consequences of aid loss than student-level factors.

8 Robustness Checks

To assess the sensitivity of my findings, I conduct a series of robustness checks focused on academic and graduation outcomes. I exclude checks for major switching and early-career earnings, where effects are limited or inconsistent, and for financial aid outcomes, which are mechanically tied to grant eligibility. The analysis is limited to the Full and Credit specifications at the second-year benchmark.

Tables 14 and 15 present results from a range of alternative specifications. Columns 1 and 2 report estimates from a regression discontinuity (RD) framework, which excludes non-recipients and removes all terms involving recipient status from Equations 3 and 4. These estimates provide a benchmark for comparison, illustrating how estimates differ when non-recipients are excluded and course failure effects are not separately accounted for.

Column 3 implements a single running variable approach using a standardized measure of distance from the two renewal thresholds. For each student, I calculate their standardized distance from both the GPA and credit completion cutoffs and use the smaller of the two as the running variable. This approach reflects the fact that students lose eligibility if they fall below either threshold, enabling estimation of the overall effect of aid loss within a unified single-dimensional diff.-in-disc. framework. The optimal bandwidth for this specification is 0.685 standard deviations, which corresponds to 3 passed credits and 0.328 GPA points. As in the main analysis, this specification includes students who are near at least one threshold.

Columns 4–7 assess robustness to alternative bandwidth choices. Columns 4 and 5 use wider bandwidths—7 credits around the credit threshold and 0.5 GPA points around the GPA threshold. I do not extend bandwidths beyond these values, as students who fail more than 8 credits or fall more than 0.5 GPA points below the threshold typically trigger SAP violations. Columns 6 and 7 use narrower bandwidths of 3 credits and 0.175 GPA points.

Columns 8 and 9 implement an overidentified instrumented difference-in-differences (DDIV) framework as an alternative to the multidimensional diff.-in-disc. approach used in the main analysis. In the Full specification, I replace the single combined instrument with two separate instruments: one for falling below the GPA threshold and one for falling below the credit threshold (each interacted with recipient status). This setup maintains flexible controls for each running variable and their interactions with recipient status. In the Credit specification, the diff.-in-disc. design shown in Equation 4 becomes a standard instrumented difference-in-differences by removing the running variable and its interactions from the model entirely.

Finally, to address differential selection into third-year persistence, I re-estimate the models using inverse-probability weights. The weights are obtained by regressing a third-year persistence indicator on the full set of student characteristics, academic controls, and fixed effects for all students close to a threshold, after which the analysis is restricted to students who persist. These results

appear in columns 10 and 11.

Panel A of Table 14 shows that the estimated effects on third-year academic outcomes are similar across specifications and closely match the main results presented in Table 5. Across all columns, estimates for full-time enrollment range from 6 to 9 pp declines in the Full specification and 17 to 24 pp declines in the Credit specification, closely aligning with the 6.5 and 23.0 pp drops in the main analysis. Similarly, estimated impacts on credits attempted and credits passed remain negative and comparable in magnitude, though some alternative specifications yield less precise estimates. Notably, the estimate on GPA in the RD specification in column 2 is negative and significant, as it reflects both the effect of aid loss and the impact of course failure. GPA estimates in the remaining columns are similar in size to the main results, providing further evidence that incorporating non-recipient students distinguishes the effect of aid loss from the academic consequences of failing a course. Estimates on fourth-year academic outcomes in Panel B are also broadly consistent with those reported in Table 5.

Specification	RD		1D Diff.-in-Disc.		Diff-in-Disc		DDIV		Weighted		
Sample	Full (1)	Credit (2)	Full (3)	Full (4)	Credit (5)	Full (6)	Credit (7)	Full (8)	Credit (9)	Full (10)	Credit (11)
Panel A: 3rd-Year Academic Outcomes											
Transfer	-0.017 (0.013)	-0.025 (0.037)	-0.001 (0.013)	-0.010 (0.012)	-0.016 (0.032)	0.016 (0.022)	0.003 (0.050)	-0.009 (0.013)	-0.024 (0.031)	0.000 (0.015)	-0.021 (0.041)
Full-Time Enrollment	-0.076*** (0.027)	-0.168** (0.067)	-0.068** (0.029)	-0.064** (0.025)	-0.208*** (0.063)	-0.091* (0.047)	-0.230** (0.100)	-0.065** (0.030)	-0.237*** (0.071)	-0.069** (0.032)	-0.240*** (0.078)
Credits Attempted	-1.032** (0.429)	-1.311 (1.040)	-0.697 (0.472)	-0.420 (0.3418)	-3.524*** (1.019)	-0.587 (0.767)	-3.624** (1.586)	-0.577 (0.510)	-3.819*** (1.179)	-0.589 (0.525)	-4.000*** (1.254)
Credits Passed	-1.501*** (0.495)	-2.473** (1.201)	-0.855 (0.543)	-0.382 (0.477)	-4.010*** (1.167)	-0.464 (0.876)	-3.640** (1.818)	-0.460 (0.542)	-4.036*** (1.241)	-0.438 (0.596)	-4.210*** (1.415)
GPA	-0.029 (0.044)	-0.582*** (0.117)	0.037 (0.047)	0.015 (0.041)	0.018 (0.112)	0.111 (0.076)	0.081 (0.176)	0.063 (0.040)	-0.015 (0.089)	0.044 (0.051)	0.065 (0.136)
N	41,198	34,184	141,946	251,547	216,162	125,894	107,220	182,137	152,585	182,137	152,585
Panel B: 4th-Year Academic Outcomes											
Persistence	-0.015 (0.016)	0.031 (0.037)	-0.045** (0.018)	-0.025 (0.017)	-0.163*** (0.039)	-0.069** (0.030)	-0.267*** (0.061)	-0.017 (0.024)	-0.180*** (0.055)	-0.034 (0.021)	-0.196*** (0.048)
Transfer	0.005 (0.015)	0.002 (0.041)	-0.045** (0.018)	0.012 (0.014)	0.024 (0.036)	0.047* (0.026)	0.036 (0.059)	0.004 (0.016)	0.006 (0.037)	0.023 (0.018)	0.007 (0.047)
Full-Time Enrollment	-0.060** (0.028)	-0.073 (0.064)	-0.043 (0.031)	-0.018 (0.027)	-0.149** (0.061)	-0.019 (0.050)	-0.224** (0.099)	-0.019 (0.034)	-0.107 (0.079)	0.007 (0.034)	-0.123 (0.075)
Credits Attempted	-0.888* (0.489)	0.308 (1.167)	-0.666 (0.546)	-0.203 (0.480)	-1.800 (1.116)	-0.243 (0.867)	-1.835 (1.787)	-0.434 (0.547)	-0.673 (1.262)	-0.021 (0.595)	-0.967 (1.357)
Credits Passed	-1.328** (0.548)	-0.986 (1.288)	-0.872 (0.608)	-0.094 (0.531)	-1.722 (1.244)	-0.205 (0.962)	-1.924 (1.962)	-0.102 (0.577)	-0.552 (1.321)	0.079 (0.659)	-0.808 (1.495)
GPA	-0.048 (0.051)	-0.372*** (0.128)	-0.012 (0.055)	0.022 (0.048)	0.208* (0.124)	0.056 (0.089)	0.216 (0.194)	0.090* (0.048)	0.213** (0.107)	0.028 (0.060)	0.266* (0.152)
N	39,399	32,235	122,384	216,063	187,037	108,748	92,927	157,982	133,220	157,982	133,220
Includes Non-Recipients?	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Credits Passed Range	≥19	[19,29]	≥21	≥17	[17,31]	≥21	[21,27]	≥19	[19,29]	≥19	[19,29]
GPA Range	≥ 2.172	≥2.5	≥2.172	≥2	≥2.5	≥2.325	≥2.5	≥2.172	≥2.5	≥2.172	≥2.5

Notes: Columns (1)-(2) report regression discontinuity (RD) estimates that exclude non-recipients. Column (3) uses a single running variable based on the minimum standardized distance to the GPA and credit thresholds. Columns (4)-(7) vary the bandwidth around each threshold. Columns (8)-(9) report alternative IV specifications. Columns (10)-(11) reweight third-year persistsers using inverse-probability weights. “GPA” only considers courses during the given year, not cumulative GPA. 4th-Year academic outcomes are conditional on students enrolling a fourth year. *p<0.10; **p<0.05; ***p<0.01.

Table 14. Academic Outcomes Robustness Check

Table 15 shows that estimates for graduation outcomes are generally robust across specifications, with a few notable differences. In the RD specification in column 2, estimates are larger and negative,

with most outcomes statistically significant. These results again underscore the importance of incorporating non-recipient students to isolate the effect of aid loss from the academic consequences of course failure. In columns 4 and 5, which use larger bandwidths, most estimates for six-year graduation rates and degree attainment become significant, with magnitudes similar to those in the main analysis but estimated more precisely. The remaining specifications yield results comparable in magnitude and significance to those reported in Table 7.

Table 15 shows that estimates for degree attainment and graduation rates are generally robust across specifications, with a few notable differences. Columns 1 and 2 yield larger negative estimates, with degree attainment becoming statistically significant in the Full specification—underscoring the importance of accounting for course failure effects through the inclusion of non-recipients in the main design. In column 3, estimates for five- and six-year graduation rates and overall degree attainment are negative and significant, suggesting that aid loss hampers timely completion regardless of which threshold is missed. The remaining specifications produce results similar in magnitude and significance to those reported in Table 7.

Specification	RD		1D Diff.-in-Disc.	Diff-in-Disc				DDIV		Weighted	
Sample	Full (1)	Credit (2)	Full (3)	Full (4)	Credit (5)	Full (6)	Credit (7)	Full (8)	Credit (9)	Full (10)	Credit (11)
Graduate in 4 Years	0.034 (0.026)	-0.083 (0.059)	0.003 (0.023)	0.046* (0.024)	0.038 (0.057)	0.041 (0.046)	0.067 (0.094)	0.059* (0.033)	0.098 (0.080)	0.055* (0.031)	0.072 (0.072)
N	41,918	34,184	141,946	251,547	216,162	125,894	107,220	182,137	152,585	182,137	152,585
Graduate in 5 Years	-0.046 (0.029)	-0.156** (0.071)	-0.071*** (0.026)	0.018 (0.027)	-0.067 (0.067)	0.047 (0.050)	-0.019 (0.104)	0.031 (0.028)	-0.001 (0.064)	0.040 (0.034)	0.003 (0.084)
N	35,796	28,928	125,041	219,340	187,245	110,394	93,295	159,618	132,634	159,618	132,634
Graduate in 6 Years	-0.040 (0.029)	-0.195*** (0.073)	-0.067*** (0.025)	-0.011 (0.027)	-0.115* (0.069)	0.007 (0.049)	-0.086 (0.104)	0.020 (0.027)	-0.030 (0.050)	0.027 (0.031)	-0.033 (0.078)
Bachelor's Degree	-0.054** (0.027)	-0.219*** (0.068)	-0.064*** (0.023)	-0.052** (0.022)	-0.141** (0.064)	-0.056 (0.044)	-0.137 (0.094)	-0.007 (0.023)	-0.110** (0.050)	-0.012 (0.031)	-0.106 (0.078)
N	29,806	23,956	106,506	185,894	157,882	93,771	78,806	135,531	112,027	135,531	112,027
Time to Graduate (Months)	-0.438 (0.859)	0.442 (2.361)	0.230 (0.763)	-1.005 (0.829)	-1.679 (2.241)	-1.991 (1.476)	-2.706 (3.208)	-0.681 (0.901)	2.752*** (0.216)	-1.323 (1.041)	-3.983 (2.830)
N	33,768	28,067	117,312	213,534	186,619	104,506	90,063	152,404	129,546	152,404	129,546
Includes Non-Recipients?	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Credits Passed Range	≥19	[19,29]	≥21	≥17	[17,31]	≥21	[21,27]	≥19	[19,29]	≥19	[19,29]
GPA Range	≥2.172	≥2.5	≥2.172	≥2	≥2.5	≥2.325	≥2.5	≥2.172	≥2.5	≥2.172	≥2.5

Notes: Columns (1)-(2) report regression discontinuity (RD) estimates that exclude non-recipients. Column (3) uses a single running variable based on the minimum standardized distance to the GPA and credit thresholds. Columns (4)-(7) vary the bandwidth around each threshold. Columns (8)-(9) report alternative IV specifications. Columns (10)-(11) reweight third-year persisters using inverse-probability weights. Four-year graduation rates include all cohorts; five-year graduation rates omit the fall 2018 entry cohort as spring 2023 graduation records are not yet available; six-year graduation rates and degree attainment further omit the fall 2017 entry cohort. Time to graduation only includes graduates, using the month and year of their graduation and assumes a uniform starting month of September for all universities. *p<0.10; **p<0.05; ***p<0.01.

Table 15. Graduation Outcomes Robustness Check

9 Conclusion

This study examines the consequences of losing financial aid after the second- and third-year renewal checkpoints of the TEXAS Grant, highlighting how both the timing and reason for aid loss shape student responses. While prior research has largely focused on first-year aid renewal, these results demonstrate that aid loss later in college poses distinct challenges with implications for persistence, academic performance, and degree completion.

Losing the TEXAS Grant leads to immediate academic disruptions, including declines in persistence and enrollment intensity. These effects are most pronounced among students who fall short of only the credit renewal requirement despite otherwise strong academic standing, indicating that aid loss disrupts students' attachment to college even when academic performance remains relatively strong. However, this drop in persistence in the immediate years following aid loss does not translate into reductions in degree attainment; instead, the pattern of results is consistent with students leaving college earlier. Importantly, these patterns are similar regardless of whether students lose aid after the second or third year, indicating that the consequences of aid loss later in college are broadly comparable across renewal checkpoints. Many students who leave following aid loss have already accumulated substantial credits and maintained relatively high GPAs, underscoring that financial constraints—rather than academic struggles—are central to the observed changes in enrollment behavior.

However, these estimates capture outcomes only after aid loss and therefore miss enrollment responses that occur earlier, when students first fail to meet renewal requirements. Students who lose eligibility after the second year are significantly less likely to persist into the third year, and a similar pattern emerges for persistence from the third to the fourth year among those who become ineligible after the third year. These responses are primarily among students who narrowly miss only the credit requirement and are accompanied by negative and significant declines in degree attainment in the reduced-form analysis. Taken together, this evidence suggests that analyses centered on observed aid loss may understate the consequences of renewal requirements, as some students respond earlier by leaving college before aid loss is realized.

This study contributes to the financial aid literature by moving beyond first-year benchmarks to examine how mid-college aid loss affects student trajectories. The findings highlight that even academically successful students remain vulnerable to financial shocks later in college, suggesting that this renewal structure may insufficiently distinguish between signs of academic struggle and indicators of continued progress toward degree completion. Together, the results underscore the importance of stable financial aid not only early in college, but throughout students' academic paths.

In addition to differences in renewal structure, the nature of the effects documented here differs from those emphasized in prior studies of financial aid loss. The existing literature focuses on merit-based programs and first-year renewal and finds relatively modest impacts on enrollment and graduation. In contrast, this study shows that aid loss later in college leads to substantial disruptions

in persistence and enrollment intensity, even among students with strong academic performance. The concentration of effects among students only below the credit requirement highlights a margin that is absent from GPA-based renewal systems and underexplored in prior work. This distinction emphasizes the importance of considering how renewal criteria interact with students' progress through their degree programs when assessing the consequences of aid loss.

While this study provides important new evidence on the consequences of financial aid loss, the findings should be interpreted with care regarding their generalizability. The TEXAS Grant program is relatively unique in its use of a credit completed renewal requirement, unlike many state and institutional aid programs that base renewal on GPA and/or credits attempted. As a result, the declines observed among students who narrowly missed the credit requirement may not extend to settings with less stringent or differently structured criteria. Within the TEXAS Grant population, the analysis focuses on students near at least one renewal threshold, representing 38.4% of second-year recipients who persisted into the third year. Students in the Credit specification account for 24.4% of all second-year recipients, indicating that the credit analysis captures a substantial and policy-relevant share of the broader recipient pool. While the findings apply most directly to this subset of students, they reflect the real-world experiences of many navigating renewal in practice.

9.1 Policy Implications

These findings carry important implications for the design of financial aid policies, especially amid ongoing concerns about college affordability. While rising tuition costs have been partially offset by expanded grant aid, many programs still impose rigid renewal criteria that may conflict with the broader goal of promoting degree completion. Given that aid loss often affects students who have already made substantial academic progress, current renewal structures may inadvertently hinder students' path to a degree.

One potential policy reform is the introduction of a probationary period following aid loss. Under such a policy, students who narrowly miss renewal requirements could either retain the TEXAS Grant temporarily during the following year, conditional on meeting both GPA and credit benchmarks, or lose aid initially but regain it if they satisfy the requirements by the end of that year. Descriptive evidence suggests that offering a path to requalification could meaningfully improve outcomes. Among the 5,156 students who lost the TEXAS Grant after their second year, 1,762 (34%) went on to meet both renewal thresholds by the end of the third year. This figure likely underestimates the potential benefits of a probationary or requalification policy, since aid loss is associated with sharp declines in full-time enrollment, making it harder for students to complete sufficient credits even if academically capable. Allowing students to maintain or regain eligibility based on demonstrated progress could help preserve financial support for those recovering from temporary setbacks and promote persistence and continued progress toward completion. While this study cannot directly estimate the effects of regaining aid, the strong negative consequences of aid loss suggest that restoring support could help mitigate some of the adverse academic and enrollment impacts documented here.

Additionally, policymakers could consider modifying the credit completion requirement to focus on credits attempted rather than a fixed number of passed credits. Currently, the TEXAS Grant requires students to pass 24 credits annually to remain eligible, irrespective of how many credits they attempt. This structure penalizes students who take full course loads but fall slightly short due to course failures, withdrawals, or other temporary setbacks. An alternative approach would align the requirement with the federal financial aid standard, which mandates that students pass at least two-thirds of their attempted credits. This shift would ensure that students who remain enrolled and continue attempting a full course load are not rendered ineligible due to setbacks in individual courses. Given that students who failed to meet the current credit threshold experienced the most severe academic and financial consequences, adopting an attempted-credit-based metric could offer greater flexibility while still promoting meaningful academic progress.

As an alternative to modifying the credit completion metric, policymakers could consider introducing a one-time course failure waiver. Under such a policy, students who attempt a full-time course load but narrowly miss the 24-credit requirement would retain TEXAS Grant eligibility. This reform would directly target the academically capable group of students who experienced the most severe disruptions in persistence and enrollment intensity following aid loss, while still maintaining incentives for continued academic progress. By allowing students to remain eligible after a temporary setback, such a waiver could reduce short-run disengagement and help preserve academic momentum during a critical stage of students' degree progression.

In terms of scope, this policy would apply to a meaningful number of TEXAS Grant recipients. Among students in the Credit specification, 1,288 students who lost the TEXAS Grant after their second year and 639 after their third year would have retained eligibility under a one-time failure waiver. This policy would have cost the state approximately \$19.68 million across 8 cohorts, or 0.65% of the state's \$400 million annual TEXAS Grant budget.²⁹ Given that the TEXAS Grant is designed to promote college access and continued progress toward a degree for low-income students, a targeted waiver of this kind may better align renewal rules with the program's objectives without requiring broad expansions of eligibility. Overall, this represents a relatively simple and modestly priced reform that could allow many students to maintain financial support at a critical stage of their academic careers.

9.2 Future Research

While this study focuses on aid loss triggered by academic renewal benchmarks, future research should examine the broader landscape of financial aid disruptions. Many students lose aid for rea-

²⁹This estimate reflects the additional TEXAS Grant disbursements that would have been paid to students who below only the credit completion requirement but remained enrolled. For students affected after the second-year renewal benchmark, the calculation uses the average number of additional years of grant receipt observed among continuing recipients (1.986 years) and the average annual award during those years (\$5,847). For students affected after the third-year benchmark, the corresponding values are 1.254 additional years and an average annual award of \$5,852. These per-student costs are multiplied by the number of students who would have retained eligibility under the proposed waiver at each benchmark and aggregated across eight cohorts. The calculation is intended to illustrate the scale of the policy rather than provide a full cost-benefit analysis.

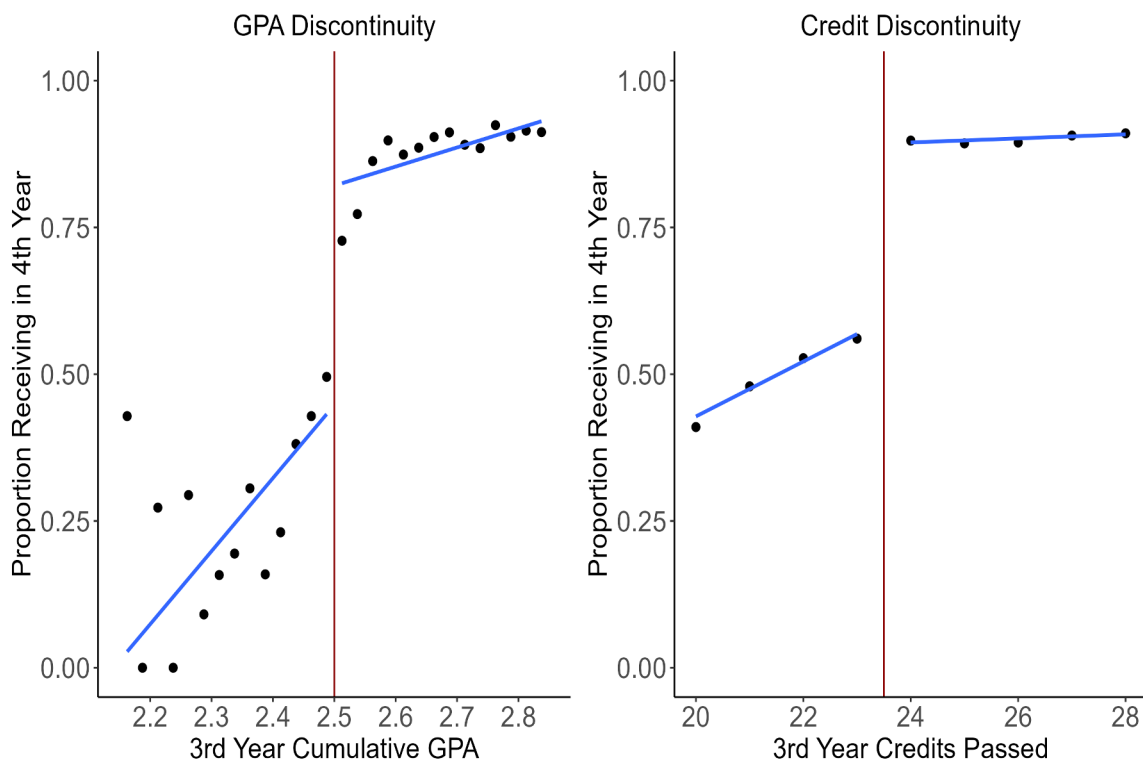
sons unrelated to academic performance, such as modest increases in family income that push them above need-based eligibility cutoffs or administrative barriers like failing to complete the FAFSA. The recent challenges surrounding FAFSA processing delays and federal funding uncertainties underscore the fragile and often unpredictable nature of financial aid systems. Understanding how such disruptions affect student outcomes is essential for designing policies that promote both accessibility and stability in aid delivery.

As financial aid programs evolve, policymakers must look beyond access alone and consider how program structures influence outcomes after initial enrollment. The evidence presented here highlights that aid loss can affect students well beyond the point of initial access. Future research should further investigate the mechanisms driving these outcomes, including students' employment decisions, borrowing behavior, and long-term labor market trajectories.

While more work is needed to understand the full range of aid disruptions, this study points to the importance of closely examining how renewal criteria shape students' experiences later in college. Renewal rules necessarily impose structure and accountability, but they also interact with students' financial circumstances, academic pathways, and institutional constraints in complex ways. Future research should focus on how renewal policies can be designed to better accommodate student progress while still maintaining clear academic standards. In particular, understanding how students navigate temporary setbacks and respond to changes in financial support remains central to evaluating the effectiveness of aid programs.

Ultimately, the purpose of financial aid is not just to open the door to college, but to support students throughout their education. Yet many programs fall short of this mission. Aid loss during the final years of college can impose unnecessary barriers. Reforming renewal policies to better accommodate the realities of student progress—while maintaining reasonable academic standards—would help ensure that financial aid fulfills its full potential.

10 Appendix



Notes: This figure shows the first-stage for each renewal threshold separately at the third-year benchmark.

Figure A.1. First Stage – Year 3 Benchmark

Outcome	Full		Credit		GPA		Diagonal	
	Compliers	Non-Compliers	Compliers	Non-Compliers	Compliers	Non-Compliers	Compliers	Non-Compliers
GPA	2.46***	2.77	2.80***	3.06	2.35***	2.41	2.34***	2.39
Credits Passed	24.80***	24.17	21.47***	21.67	27.43**	27.75	21.33	21.43
Age	18.04	18.02	18.03	18.01	18.04	18.04	18.04	18.07
Male	0.44***	0.37	0.44***	0.36	0.44**	0.39	0.47**	0.34
White	0.16**	0.19	0.18*	0.20	0.16	0.17	0.16	0.14
Black	0.24***	0.16	0.17**	0.13	0.28***	0.19	0.23	0.23
Hispanic/Latino	0.53***	0.59	0.58	0.59	0.50***	0.59	0.55	0.54
First-Generation	0.28	0.29	0.29	0.27	0.28	0.30	0.28***	0.47
N	5,156	1,764	1,286	965	2904	729	966	70

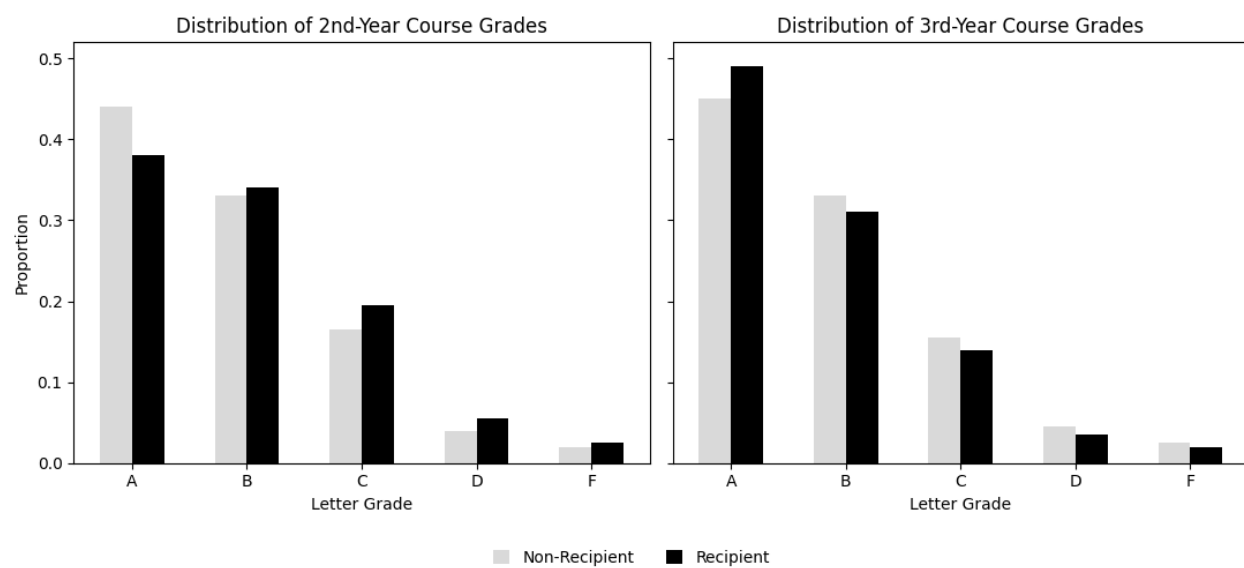
Notes: Compliers are students who lost the TEXAS Grant after falling below a renewal threshold; non-compliers retained the grant despite being below a threshold. Asteriks denote significance levels from t-tests comparing compliers and non-compliers within each specification. *p<0.10; **p<0.05; ***p<0.01.

Table A.1. Characteristics of Compliers and Non-Compliers by Specification – Year 2 Benchmark

Panel A: Student Demographics		Panel B: Family Income		Panel C: 2nd-Year Academic Outcomes	
Outcome	Estimate	Outcome	Estimate	Outcome	Estimate
Age	0.021 (0.0239)	Unknown	-0.014 (0.0268)	STEM Major	0.035 (0.0257)
Male	0.005 (0.0280)	<\$20,000	0.016 (0.0125)	Cumulative GPA	0.066*** (0.0082)
White	0.068** (0.0269)	\$20k–\$39,999	0.014 (0.0155)	Credits Attempted	0.258 (0.2132)
Black	-0.011 (0.0157)	\$40k–\$59,999	-0.014 (0.0159)	Credits Passed	0.591*** (0.2238)
Hispanic/Latino	-0.047* (0.0247)	\$60k–\$79,999	-0.004 (0.0151)	Math Credits Attempted	0.167 (0.1686)
First-Generation	-0.044** (0.0182)	\$80,000+	0.002 (0.0252)	Science Credits Attempted	0.233 (0.3331)
N	115,399	N	115,399	N	115,399

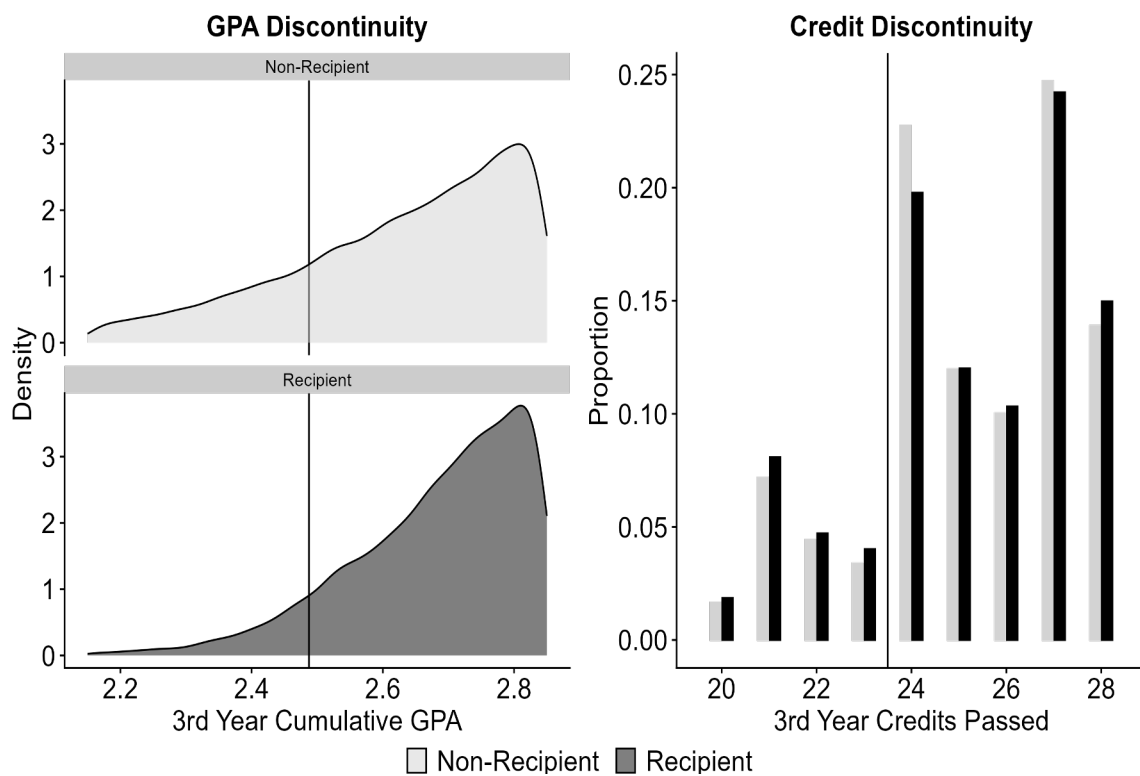
Notes: The family income bins in Panel B correspond to the options students can select on their applications. STEM majors include fields related to engineering, biological sciences, mathematics/statistics, and physical sciences. Mathematics and science courses are classified based on subject codes, with mathematics/statistics representing math courses and biology, chemistry, physics, physical sciences, geology, and general science representing science courses. *p<0.10; **p<0.05; ***p<0.01.

Table A.2. Covariate Balance Check – Year 3 Benchmark



Notes: This figure shows the distribution of course grades for recipients and non-recipients at the second-year benchmark (left panel) and third-year benchmark (right panel). Grades are coded as A, B, C, D, and F, where D is the lowest passing grade and F indicates course failure.

Figure A.2. Distribution of Course Letter Grades



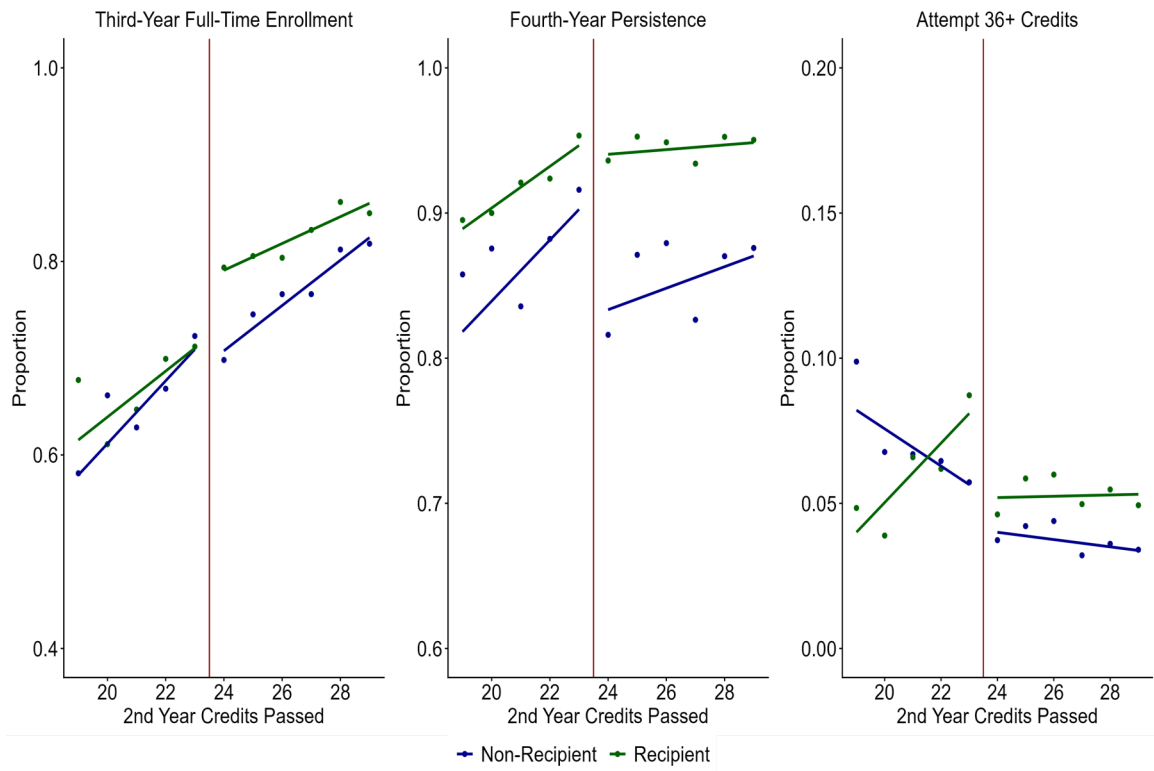
Notes: This figure shows the distribution of cumulative GPA (left panel) and credits passed (right panel) at the third-year renewal benchmark for TEXAS Grant recipients and non-recipients. The vertical lines indicate the relevant GPA (2.5) and credit (24) thresholds.

Figure A.3. Density Plots – Year 3 Benchmark

Aid Source	Estimate	Year 2 Recipient Mean
Panel A: Federal Need-Based		
Pell Grant	-311 (197)	5,452
SEOG	110*** (31)	67
Categorical Aid	-80 (115)	381
Panel B: State Need-Based		
HB3015 Grants	330** (137)	824
HB3015 Other	-12 (13)	1
TPEG	201** (88)	491
Panel C: Other Need-Based / Exemptions		
Student Deposit Fund Scholarship	14 (11)	9
Other Grants	1452*** (357)	773
Tuition Exemptions and Waivers	-148 (206)	347
Panel D: Merit-Based		
Texas Top 10% Scholarship	25*** (6)	37
Institutional Merit Aid	-618*** (212)	813
N	182,137	41,926

Notes: This table shows estimated effects on non-TEXAS Grant aid sources for students in the Full specification. “Other Grants” includes all other forms of grants not itemized in the above table. “Year 2 Recipient Mean” refers to the average amount received by TEXAS Grant recipients in Year 2. *p<0.10; **p<0.05; ***p<0.01.

Table A.3. Other Sources of Gift Aid – Year 2 Benchmark



Notes: This figure plots binned means of the outcomes in Table 6 against second-year credits passed, separately for TEXAS Grant recipients and non-recipients. Solid lines show local linear fits estimated on either side of the 24-credit renewal threshold, indicated by the vertical red line. The sample is restricted to students within the optimal bandwidth of the credit cutoff.

Figure A.4. Reduced Form Plots

Outcome	Specification			
	Full	Credit	GPA	Diagonal
Graduate in 4 Years	0.086** (0.035)	0.236 (0.080)	0.044 (0.058)	-0.166 (0.125)
Graduate in 5 Years	0.042 (0.037)	0.027 (0.093)	0.024 (0.060)	-0.213 (0.176)
Graduate in 6 Years	0.029 (0.034)	-0.016 (0.087)	0.055 (0.054)	-0.292* (0.175)
Bachelor's Degree	-0.011 (0.031)	-0.097 (0.079)	0.025 (0.049)	-0.195 (0.167)
N	135,531	112,027	39,076	20,802
Time to Graduate (Months)	-2.306* (1.185)	-7.257** (3.480)	-0.726 (1.867)	5.407 (5.937)
N	120,583	101,036	33,338	17,577

Notes: All graduation rate outcomes include cohorts entering in fall 2016 or earlier, holding the set of cohorts fixed for comparability across outcomes. Time to graduation only includes graduates of these cohorts, using the month and year of their graduation and assumes a uniform starting month of September for all universities. *p<0.10; **p<0.05; ***p<0.01.

Table A.4. Graduation Outcomes – Year 2 Benchmark, Static Cohorts

Specification	After 1 Year				After 2 Years				To Graduation			
	Major Change	Out of CIP2	To STEM	From STEM	Major Change	Out of CIP2	To STEM	From STEM	Major Change	Out of CIP2	To STEM	From STEM
Full	0.016 (0.033)	-0.026 (0.049)	-0.040 (0.031)	-0.023 (0.041)	0.008 (0.035)	-0.032 (0.038)	-0.036* (0.021)	-0.011 (0.034)	0.023 (0.040)	-0.007 (0.020)	-0.023 (0.027)	0.025 (0.039)
N	182,137	51,094	51,094	51,094	157,982	73,717	73,717	73,717	155,757	54,066	54,066	54,066
Credit	0.006 (0.080)	-0.197* (0.115)	0.043 (0.084)	-0.198** (0.095)	-0.043 (0.080)	-0.107 (0.088)	0.012 (0.052)	-0.039 (0.083)	-0.083 (0.106)	-0.051 (0.030)	-0.096 (0.069)	-0.085 (0.094)
N	152,585	41,509	41,509	41,509	133,220	61,466	61,466	61,466	132,462	44,348	44,348	44,348
GPA	0.080 (0.053)	0.060 (0.080)	-0.128*** (0.048)	0.049 (0.065)	-0.001 (0.057)	0.059 (0.065)	-0.078** (0.034)	0.006 (0.054)	0.050 (0.063)	0.000 (0.034)	-0.045 (0.043)	0.047 (0.061)
N	49,233	15,655	15,655	15,655	42,373	20,646	20,646	20,646	39,774	16,378	16,378	16,378
Diagonal	-0.072 (0.151)	0.075 (0.226)	0.082 (0.142)	-0.073 (0.205)	0.156 (0.153)	-0.053 (0.156)	-0.136* (0.073)	-0.169 (0.153)	0.047 (0.181)	-0.011 (0.051)	0.057 (0.119)	-0.054 (0.183)
N	26,592	8,648	8,648	8,648	23,479	11,589	11,589	11,589	20,559	8,784	8,784	8,784

Notes: “Major Change” is an indicator variable equal to 1 if a student’s major at the end of the third year, fourth year, or at graduation differs from their major at the end of their second year, which was the last recorded major before the renewal checkpoint. “Out of CIP2”, “To STEM”, and “From STEM” are conditional on a student changing majors. “Out of CIP2” is an indicator for switching to a major outside the 2-digit CIP category of a student’s major at the end of their second year. “To STEM” indicates that a student’s major at the end of the second year was not in STEM, but their major at the end of the third year, fourth year, or at graduation was in STEM. “From STEM” is defined analogously. *p<0.10; **p<0.05; ***p<0.01.

Table A.5. Major Changes – Year 2 Benchmark

Specification	All			Graduates		
	1 Year	2 Years	3 Years	1 Year	2 Years	3 Years
Full	957 (1810)	1501 (2335)	-1734 (2819)	113 (2054)	721 (2624)	-1391 (3216)
N	114,604	92,856	75,069	104,711	85,055	68,705
Credit	-1828 (5018)	-865 (6714)	835 (7738)	-3012 (6003)	-7839 (8440)	-1296 (9553)
N	94,345	75,914	60,945	87,440	70,506	56,584
GPA	-456 (2715)	1635 (3546)	-4809 (4400)	-1749 (3020)	1327 (3877)	-5700 (4961)
N	33,627	27,956	23,289	29,369	24,633	20,524
Diagonal	4416 (7514)	1700 (9788)	-14002 (11575)	9414 (8954)	2588 (10843)	-6562 (13279)
N	17,770	14,604	12,133	15,080	12,538	10,413

Notes: Earnings are calculated starting with the first full calendar year after a student is no longer enrolled. Individuals with no earnings reported to the TWC or are still enrolled are omitted from this analysis. I cannot determine whether individuals with no reported earnings are unemployed, have non-reported earnings, or are no longer in Texas. “All” includes all students who meet this requirement, while “Graduates” restricts to those who earned a bachelor’s degree. *p<0.10; **p<0.05; ***p<0.01.

Table A.6. Early-Career Earnings – Year 2 Benchmark

Outcome	Specification			
	Full	Credit	GPA	Diagonal
Graduate in 4 Years	-0.078 (0.067)	0.081 (0.123)	-0.127 (0.118)	-0.233 (0.241)
Graduate in 5 Years	-0.121* (0.072)	-0.110 (0.144)	-0.031 (0.114)	-0.224 (0.372)
Graduate in 6 Years	-0.068 (0.064)	-0.076 (0.128)	-0.012 (0.097)	-0.208 (0.392)
Bachelor's Degree	-0.012 (0.056)	-0.121 (0.112)	-0.056 (0.085)	-0.336 (0.361)
N	85,435	72,775	20,770	9,922
Time to Graduate (Months)	3.135 (2.268)	2.167 (4.358)	1.557 (3.674)	0.554 (11.273)
N	80,869	69,395	19,023	8,875

Notes: All graduation rate outcomes include cohorts entering in fall 2016 or earlier, holding the set of cohorts fixed for comparability across outcomes. Time to graduation only includes graduates of these cohorts, using the month and year of their graduation and assumes a uniform starting month of September for all universities. *p<0.10; **p<0.05; ***p<0.01.

Table A.7. Graduation Outcomes – Year 3 Benchmark, Static Cohorts

Specification	After 1 Year				To Graduation			
	Major Change	Out of CIP2	To STEM	From STEM	Major Change	Out of CIP2	To STEM	From STEM
Full	0.009 (0.056)	0.085 (0.093)	0.011 (0.062)	-0.069 (0.110)	0.173** (0.082)	0.018 (0.066)	-0.036 (0.063)	-0.003 (0.074)
N	115,399	33,732	33,732	33,732	104,613	31,031	31,031	31,031
Credit	0.210* (0.114)	-0.013 (0.164)	-0.065 (0.106)	-0.094 (0.215)	0.297* (0.167)	-0.031 (0.124)	-0.144 (0.120)	0.202 (0.174)
N	99,984	29,554	29,554	29,554	91,214	25,560	25,560	25,560
GPA	-0.093 (0.083)	-0.027 (0.131)	0.066 (0.089)	-0.122 (0.128)	0.263** (0.122)	0.113 (0.099)	-0.051 (0.098)	-0.190** (0.095)
N	25,422	6,668	6,668	6,668	22,140	8,768	8,768	8,768
Diagonal	0.040 (0.365)	-0.182 (0.576)	-0.160 (0.799)	0.040 (0.620)	-0.421 (0.446)	0.066 (0.490)	0.005 (0.473)	0.361 (0.525)
N	12,291	3,266	3,266	3,266	10,195	4,001	4,001	4,001

Notes: “Major Change” is an indicator variable equal to 1 if a student’s major at the end of the fourth year or at graduation differs from their major at the end of their third year, which was the last recorded major before the renewal checkpoint. “Out of CIP2”, “To STEM”, and “From STEM” are conditional on a student changing majors. “Out of CIP2” is an indicator for switching to a major outside the 2-digit CIP category of a student’s major at the end of their third year. “To STEM” indicates that a student’s major at the end of the third year was not in STEM, but their major at the end of the fourth year or at graduation was in STEM. “From STEM” is defined analogously. *p<0.10; **p<0.05; ***p<0.01.

Table A.8. Major Changes – Year 3 Benchmark

Specification	All			Graduates		
	1 Year	2 Years	3 Years	1 Year	2 Years	3 Years
Full	4078 (3840)	1922 (4411)	2377 (6338)	3053 (4065)	697 (4746)	1718 (6868)
N	69,609	55,830	43,941	67,013	53,908	42,423
Credit	20247** (8787)	18609* (9683)	13968 (14618)	12919 (8951)	11015 (9987)	1436 (15835)
N	59,528	47,380	36,952	57,683	46,064	35,923
GPA	1074 (5484)	2466 (6561)	10304 (9969)	3210 (5876)	2869 (7079)	14652 (10528)
N	17,465	14,551	11,919	16,280	13,639	11,181
Diagonal	-22909 (16866)	-27015* (15790)	14105 (39072)	-20743 (20107)	-30119* (16434)	24988 (47250)
N	8,234	6,785	5,487	7,486	6,225	5,048

Notes: Earnings are calculated starting with the first full calendar year after a student is no longer enrolled. Individuals with no earnings reported to the TWC or are still enrolled are omitted from this analysis. I cannot determine whether individuals with no reported earnings are unemployed, have non-reported earnings, or are no longer in Texas. “All” includes all students who meet this requirement, while “Graduates” restricts to those who earned a bachelor’s degree. *p<0.10; **p<0.05; ***p<0.01.

Table A.9. Early Career Earnings – Year 3 Benchmark

	Hispanic/Latino		White	
Specification	Full	Credit	Full	Credit
Panel A: 3rd-Year Academic Outcomes				
Full-Time Enrollment	-0.085 (0.053)	-0.314*** (0.117)	-0.007 (0.070)	-0.058 (0.177)
Credits Attempted	-0.712 (0.866)	-3.771** (1.900)	0.761 (1.163)	-2.596 (2.921)
Attempt 36+ Credits	0.019 (0.033)	-0.018 (0.065)	0.024 (0.051)	-0.209* (0.126)
Panel B: Graduation Outcomes				
Graduate in 4 Years	0.073 (0.050)	0.013 (0.105)	0.091 (0.075)	0.161 (0.185)
N	59,382	49,186	82,780	71,150
Bachelor's Degree	0.046 (0.050)	0.033 (0.120)	-0.042 (0.067)	-0.146 (0.167)
N	42,355	34,503	63,620	54,086
Time to Graduate (Months)	-2.749 (1.798)	-3.856 (4.480)	-3.651 (2.265)	-12.874* (6.591)
N	48,219	40,533	71,682	62,234

Notes: Four-year graduation rates include all cohorts; degree attainment further omit the fall 2017 and 2018 entry cohorts. Time to graduation only includes graduates, using the month and year of their graduation and assumes a uniform starting month of September for all universities. *p<0.10; **p<0.05; ***p<0.01.

Table A.10. Academic and Graduation Outcomes, By Race/Ethnicity – Year 2 Benchmark

	Non-First-Generation		First-Generation	
Specification	Full	Credit	Full	Credit
Panel A: 3rd-Year Academic Outcomes				
Full-Time Enrollment	-0.072 (0.047)	-0.281* (0.160)	-0.143* (0.072)	-0.470*** (0.172)
Credits Attempted	-0.084 (0.770)	-2.940 (2.542)	-2.901** (1.190)	-13.164*** (3.144)
Attempt 36+ Credits	0.007 (0.033)	-0.088 (0.103)	-0.073 (0.044)	-0.287*** (0.105)
N	118,097	100,518	26,315	21,547
Panel B: Graduation Outcomes				
Graduate in 4 Years	0.045 (0.047)	0.317* (0.167)	0.054 (0.071)	-0.291* (0.156)
N	118,097	100,518	26,315	21,547
Bachelor's Degree	0.031 (0.046)	0.016 (0.173)	-0.055 (0.066)	-0.313** (0.130)
N	84,397	70,983	19,180	15,525
Time to Graduate (Months)	0.140 (1.476)	-13.372** (6.455)	-3.683 (2.336)	4.431 (5.866)
N	100,060	86,334	21,477	17,907

Notes: Four-year graduation rates include all cohorts; degree attainment further omit the fall 2017 and 2018 entry cohorts. Time to graduation only includes graduates, using the month and year of their graduation and assumes a uniform starting month of September for all universities. *p<0.10; **p<0.05; ***p<0.01.

Table A.11. Academic and Graduation Outcomes, By First-Generation Status – Year 2 Benchmark

References

- Angrist, J., D. Autor, and A. Pallais (2022). Marginal effects of merit aid for low-income students. *The Quarterly Journal of Economics* 137(2), 1039–1090.
- Baker, A. and C. Montalto (2019). Student loan debt and financial stress: Implications for academic performance. *Journal of College Student Development* 60(1), 115–120.
- Barreca, A., J. Lindo, and G. Waddell (2016). Heaping-induced bias in regression-discontinuity designs. *Economic inquiry* 54(1), 268–293.
- Bettinger, E. (2004). How financial aid affects persistence. In *College Choices: The Economics of Where to Go, When to Go, and How to Pay For it*, NBER Chapters, pp. 207–238. National Bureau of Economic Research.
- Bettinger, E., B. Long, P. Oreopoulos, and L. Sanbonmatsu (2012). The role of application assistance and information in college decisions: Results from the h&r block fafsa experiment. *The Quarterly Journal of Economics* 127(3), 1205–1242.
- Calonico, S., M. Cattaneo, and R. Titiunik (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica* 82(6), 2295–2326.
- Carruthers, C. and U. Özek (2016). Losing HOPE: Financial aid and the line between college and work. *Economics of Education Review* 53, 1–15.
- Castleman, B. and B. Long (2016). Looking beyond enrollment: The causal effect of need-based grants on college access, persistence, and graduation. *Journal of Labor Economics* 34(4), 1023–1073.
- Castleman, B. and L. Page (2016). Freshman year financial aid nudges: An experiment to increase fafsa renewal and college persistence. *Journal of Human Resources* 51(2), 389–415.
- Choi, J. and M. Lee (2018). Regression discontinuity with multiple running variables allowing partial effects. *Political Analysis* 26(3), 258–274.
- Columbus, R., K. Cummings, K. Deane, J. Skiles, S. DesJardins, and B. McCall (2025). Examining how oncome influences the effects of losing merit aid among Tennessee community college students. *AERA Open* 11.
- Cummings, K., K. Deane, B. McCall, and S. DesJardins (2022). Exploring race and income heterogeneity in the effects of state merit aid loss among four-year college entrants. *The Journal of Higher Education* 93(6), 873–900.
- Denning, J. (2019). Born under a lucky star: Financial aid, college completion, labor supply, and credit constraints. *Journal of Human Resources* 54(3), 760–784.
- Denning, J., B. Marx, and L. Turner (2019). Propelled: The effects of grants on graduation, earnings, and welfare. *American Economic Journal: Applied Economics* 11(3), 193–224.
- Department of Homeland Security (2023). DHS STEM Designated Degree Program List. Technical report, Department of Homeland Security.

- Despard, M., D. Perantie, S. Taylor, M. Grinstein-Weiss, T. Friedline, and R. Raghavan (2016). Student debt and hardship: Evidence from a large sample of low- and moderate-income households. *Children and Youth Services Review* 70, 8–18.
- Destin, M. and R. Svoboda (2018). Costs on the mind: The influence of the financial burden of college on academic performance and cognitive functioning. *Research in Higher Education* 59, 302–324.
- Dynarski, S., C. Libassi, K. Michelmore, and S. Owen (2021). Closing the gap: The effect of reducing complexity and uncertainty in college pricing on the choices of low-income students. *American Economic Review* 111(6), 1721–1756.
- Espinosa, L., J. Turk, M. Taylor, and H. Chessman (2019). Race and ethnicity in higher education: A status report.
- Galperin, M. (2024). *Who Benefits from College Grant Aid and Why? Evidence from Texas*. Ph. D. thesis, The University of Chicago.
- Georgia Office of Research and Policy Analysis (2024). Keeping the HOPE scholarship through college: The status of fall 2018 first-time freshmen six years later.
- Goldrick-Rab, S., R. Kelchen, D. Harris, and J. Benson (2016). Reducing income inequality in educational attainment: Experimental evidence on the impact of financial aid on college completion. *American Journal of Sociology* 121(6), 1762–1817.
- Hampole, M. (2023). Financial Frictions and Human Capital Investments. *PhD Dissertation*.
- Henry, G., R. Rubenstein, and D. Bugler (2004). Is HOPE enough? Impacts of receiving and losing merit-based financial aid. *Educational Policy* 18(5), 686–709.
- Herzog, S. (2018). Financial aid and college persistence: Do student loans help or hurt? *Research in Higher Education* 59, 273–301.
- Hojman, D., A. Miranda, and J. Ruiz-Tagle (2016). Debt trajectories and mental health. *Social Science & Medicine* 167, 54–62.
- Indiana Commission for Higher Education (2025). Frank O’Bannon Grant. <https://www.in.gov/che/state-financial-aid/state-financial-aid-by-program/frank-obannon-grant/>.
- Jones, T., D. Kreisman, R. Rubenstein, C. Searcy, and R. Bhatt (2021). The effects of financial aid loss on persistence and graduation: A multi-dimensional regression discontinuity approach. *Education Finance and Policy* 17(2), 206–231.
- Kahneman, D., J. Knetsch, and R. Thaler (1991). Anomalies: The endowment effect, loss aversion, and status quo bias. *Journal of Economic perspectives* 5(1), 193–206.
- Ko, K., K. Bartoszuk, S. A. Peek, and M. Hurley (2025). Profiles of first-generation college students: Social, financial, academic, and cultural barriers to college lives. *Journal of College Student Retention: Research, Theory & Practice* 27(2), 534–554.
- Kofoed, M. (2017). To apply or not to apply: FAFSA completion and financial aid gaps. *Research in Higher Education* 58, 1–39.

- Ma, J., M. Pender, and M. Oster (2024). Trends in college price and student aid 2024. *College Board*.
- Mayer, A., R. Patel, T. Rudd, and A. Ratledge (2015). Designing scholarships to improve college success: Final report on the performance-based scholarship demonstration. *New York: MDRC (2015)*.
- Missouri Coordinating Board for Higher Education (2020). FY 2022 department operating and student financial aid budget recommendations. <https://dhewd.mo.gov/media/pdf/tab-19-september-15-2020>.
- Monks, J. (2009). The impact of merit-based financial aid on college enrollment: A field experiment. *Economics of Education Review* 28(1), 99–106.
- Montenegro, S. (2020). *Three essays on the economics of financial aid*. Ph. D. thesis, The University of Texas at Dallas.
- Nguyen, T., J. Kramer, and B. Evans (2019). The effects of grant aid on student persistence and degree attainment: A systematic review and meta-analysis of the causal evidence. *Review of educational research* 89(6), 831–874.
- Office of Management and Budget (2025). OMB M-25-13.
- Pascarella, E., C. Pierson, G. Wolniak, and P. Terenzini (2004). First-generation college students: Additional evidence on college experiences and outcomes. *The journal of higher education* 75(3), 249–284.
- Perry, R., S. Hladkyj, R. Pekrun, R. Clifton, and J. Chipperfield (2005). Perceived academic control and failure in college students: A three-year study of scholastic attainment. *Research in Higher Education* 46(5), 535–569.
- Rothstein, J. and C. Rouse (2011). Constrained after college: Student loans and early-career occupational choices. *Journal of Public Economics* 95(1-2), 149–163.
- Schudde, L. and J. Scott-Clayton (2016). Pell Grants as Performance-Based Aid? An Examination of Satisfactory Academic Progress Requirements in the Nation’s Largest Need-Based Aid Program. *Research in Higher Education* 57, 943–967.
- Selenko, E. and B. Batinic (2011). Beyond debt. A moderator analysis of the relationship between perceived financial strain and mental health. *Social Science & Medicine* 73(12), 1725–1732.
- Sjoquist, D. and J. Winters (2015). State merit-based financial aid programs and college attainment. *Journal of Regional Science* 55(3), 364–390.
- Stater, M. (2011). Financial aid, student background, and the choice of first-year college major. *Eastern Economic Journal* 37(3), 321–343.
- Stinebrickner, T. and R. Stinebrickner (2012). Learning about academic ability and the college dropout decision. *Journal of Labor Economics* 30(4), 707–748.
- Sullivan, L., T. Meschede, L. Dietrich, and T. Shapiro (2015). The racial wealth gap. *Institute for Assets and Social Policy, Brandeis University. DEMOS*.

- Sweet, E., A. Nandi, E. Adam, and T. McDade (2013). The high price of debt: Household financial debt and its impact on mental and physical health. *Social Science & Medicine* 91, 94–100.
- Texas Legislature (2023). Texas education code § 54.051(c). *Texas Statutes*.
- Unverferth, A. R., C. Talbert-Johnson, and T. Bogard (2012). Perceived barriers for first-generation students: Reforms to level the terrain. *International Journal of Educational Reform* 21(4), 238–252.
- U.S. Department of Education, National Center for Education Statistics (2023). Integrated Post-secondary Education Data System (IPEDS), student financial aid component final data (2001-02 - 2021-22) and provisional data (2022-23).
- Villarreal, M. U. (2018). *The politics and policies of higher education in Texas: 1995-2013*. Ph. D. thesis, The University of Texas at Austin.
- Walsemann, K., G. Gee, and D. Gentile (2015). Sick of our loans: Student borrowing and the mental health of young adults in the United States. *Social Science & Medicine* 124, 85–93.